Hawai‘i Department of Education

Hawai‘i Common Core Standards for Mathematics

July 17, 2026

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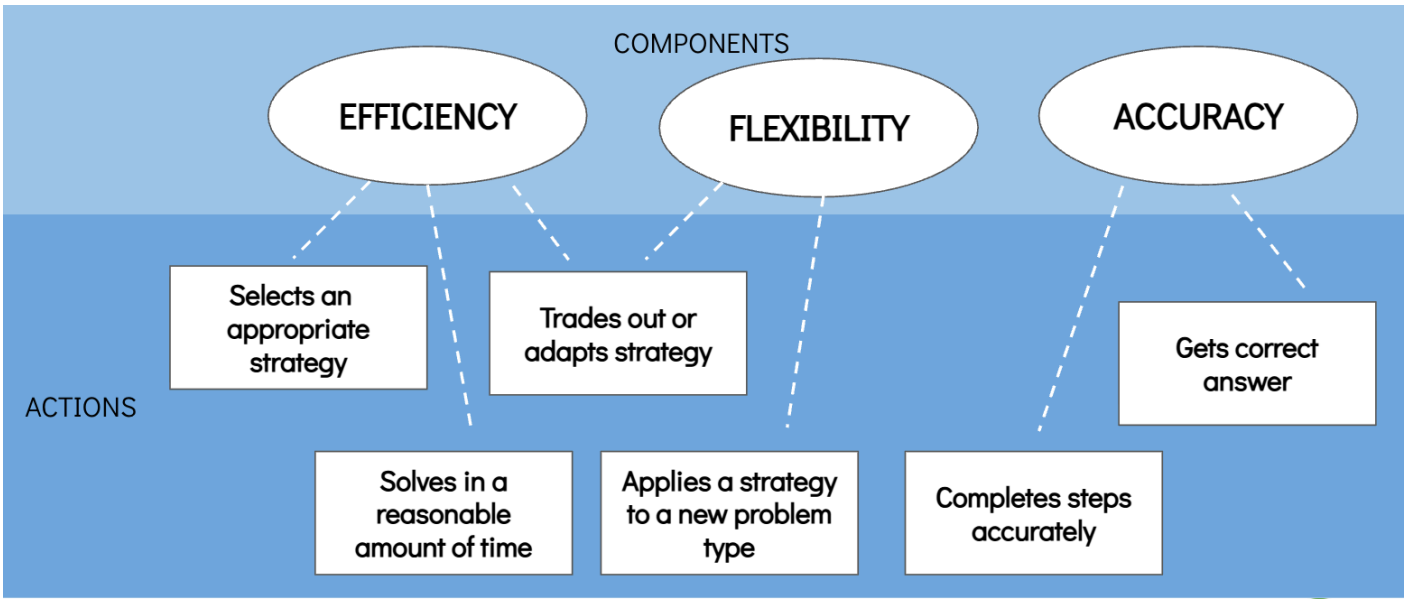
Hawai‘i Department of Education’s Commitment to Mathematical Excellence

A PreK-20 mathematics education that cultivates a joyful and inclusive learning environment that celebrates the truly diverse people of the islands remains at the heart of the work of the Hawai‘i Department of Education (Department). The Department’s revised standards are designed to build upon this foundation and strengthen the rigorous problem-solving and critical thinking skills that ensure a strong connection between mathematical concepts and real-life applications.

Through access to transformative experiences rooted in Hawai‘i’s unique communities, the Department’s goal is to empower all learners to excel and develop a lifelong appreciation for mathematics that will unlock their full potential for a brighter future. This enhanced approach is defined by a genuine mathematical rigor that requires students to:

- Apply concepts rather than memorize procedures or mimic algorithms.
- Productively struggle through tasks.
- Communicate and explain their mathematical thinking with accuracy and precision, using mathematical language and reasoning.
- Connect and build upon mathematical ideas.

The Pillars of Fluency



Adapted from Bay Williams, J. & SanGiovanni, J. (2021). *Figuring out fluency in mathematics teaching and learning: Moving beyond basics facts and memorization*. Corwin.

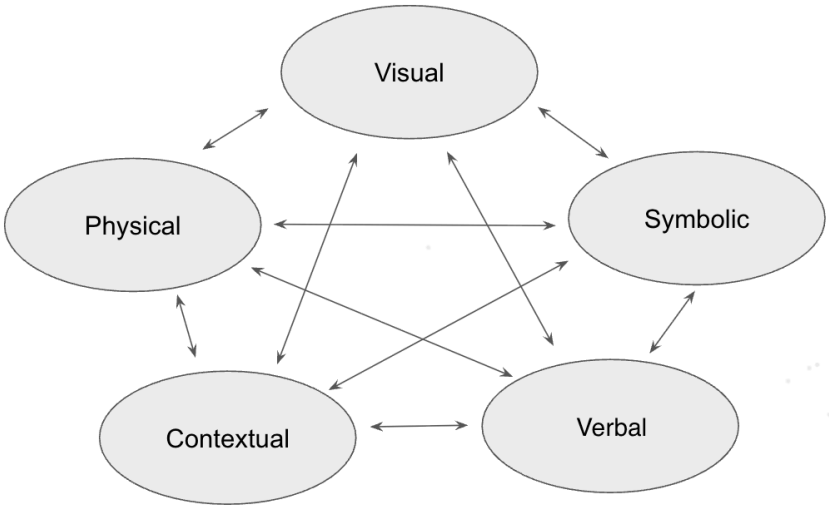
The Department defines mathematical fluency as accuracy, efficiency, and flexibility—aligning with state, national, and National Council of Teachers of Mathematics (NCTM) standards. This definition goes beyond speed and memorization, emphasizing conceptual understanding and critical thinking.

- **Accuracy** – Solving a problem correctly.
- **Efficiency** – Solving a [problem] in a reasonable amount of time by selecting an appropriate strategy and readily implementing that strategy. (Figuring Out Fluency in Mathematics Teaching and Learning: Moving Beyond Basic Facts and Memorization)
- **Flexibility** – Seeing multiple ways to solve a problem and adapting strategies as needed.

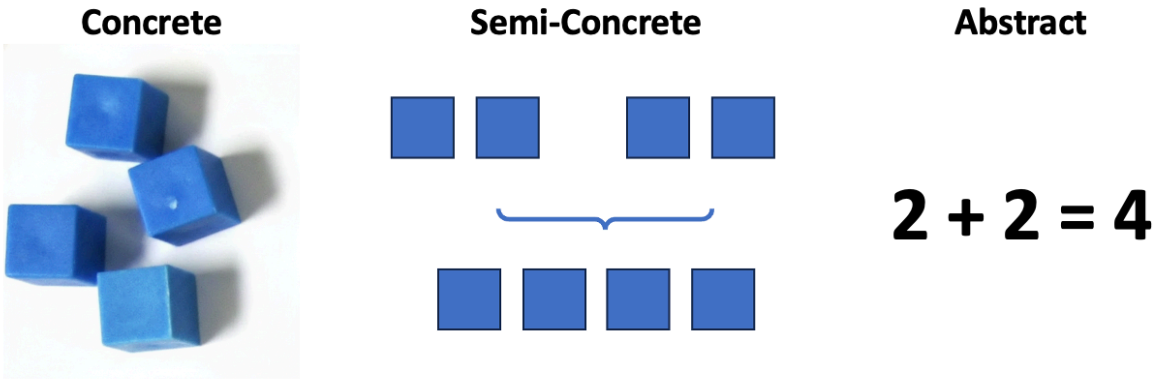
Reasoning: The Scalable Core

Fluency is built on the principle that reasoning is the core of mathematical understanding. Reasoning from foundational properties builds skills that transfer to any context whereas procedural gimmicks work for only one type of problem. Students strengthen this understanding by connecting multiple representations—using manipulatives like base ten blocks or algebra tiles—for example, to see how the distributive property applies across situations. Whether a child decomposes 7×8 into $(7 \times 5) + (7 \times 3)$ or an algebra student factors $2x^2 + 7x$ into $x(2x + 7)$, reasoning creates durable knowledge that scales across grades. Focusing on reasoning prepares students for future success and makes math feel meaningful.

Representations, as shown below, are crucial for supporting this level of understanding. For example, by building connections between concrete, semi-concrete, and abstract representations simultaneously, students see how these different forms link together all at once. This integrated approach allows them to move flexibly between ideas, ensuring they develop a deeper understanding of mathematical concepts and stronger problem-solving skills.



Adapted from Lesh, R., Post, T., & Behr, M. (1987). Representations and Translations among Representations in Mathematics Learning and Problem Solving. In C. Janvier (Ed.), *Problems of Representation in the Teaching and Learning of Mathematics* (pp. 33-40). Hillsdale, NJ: Erlbaum.



Adapted from Miller, S. P., Mercer, C.D., & Dillon, A. S. (1992). CSA: Acquiring and Retaining Math Skills.*Intervention in School and Clinic*, 28, 105-110.

A Framework for Practice and Instruction

The Standards for Mathematical Practice (SMPs) define the behaviors and habits of mind we expect in every mathematically proficient student. They show what it looks like to actively engage in mathematics. The Department's standards support these key practices:

1. Make sense of problems and persevere in solving them
2. Reason abstractly and quantitatively
3. Construct viable arguments and critique the reasoning of others
4. Model with mathematics
5. Use tools strategically
6. Attend to precision
7. Look for and make use of structure
8. Look for and express regularity in repeated reasoning

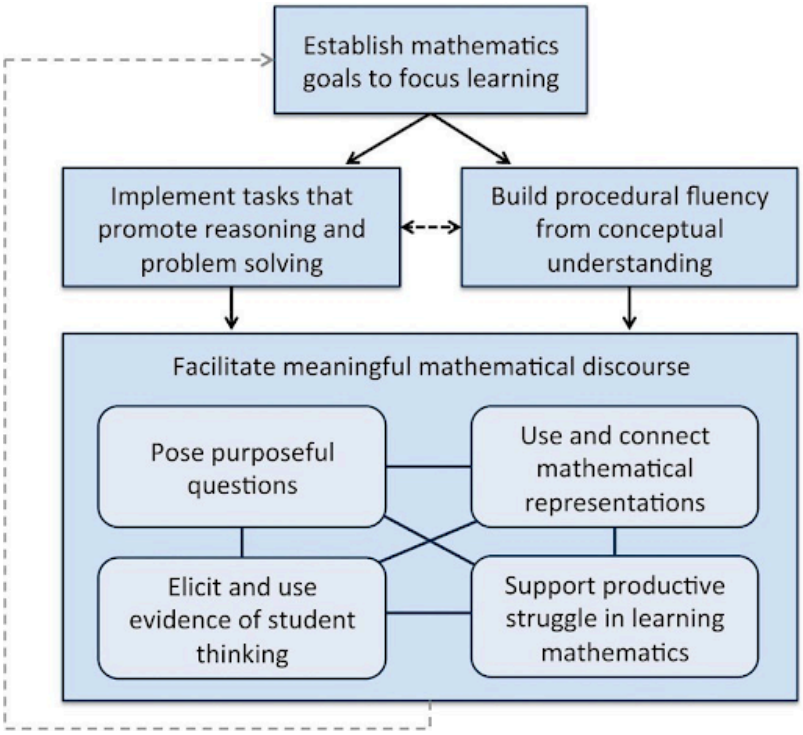
These practices are essential threads in every learning experience, ensuring mathematics is meaningful, connected, and applicable.

The Teacher's Role: The Effective Mathematics Teaching Practices (EMTPs)

Teachers are central to creating a joyful, inclusive learning environment where every student thrives. The Effective Mathematics Teaching Practices (EMTPs) provide a research-based framework of high-impact instructional moves that promote deep learning, build mathematical fluency, and support rigor. The Department's revised standards are designed to be enacted through these practices (National Council of Teachers of Mathematics, 2014):

- Establish mathematics goals to focus learning
- Implement tasks that promote reasoning and problem solving
- Use and connect mathematical representations
- Facilitate meaningful mathematical discourse

- Pose purposeful questions
- Build procedural fluency from conceptual understanding
- Support productive struggle in learning mathematics
- Elicit and use evidence of student thinking



By committing to these practices, educators ensure that Hawai‘i classrooms develop not just factual knowledge, but skilled, confident mathematical thinkers. The alignment of SMPs (what students do) and EMTPs (what teachers do) forms the foundation of the Department’s shared effort to improve mathematics education, ensuring a robust and engaging experience for both teachers and students in their mathematical journey.

Using the Standards

The Hawaii Mathematics Standards use a structured coding scheme that includes the following elements:

1. **Grade Level**
   This element identifies the grade level or high school course of the standard:
   - Kindergarten is represented by “K”
   - Grades 1-8 are represented by their respective number
   - High School courses are represented by abbreviation (e.g., “A1” for Algebra 1)

2. **Domain / ‘Ōpa‘a**
   Each standard is categorized within a domain and represented by a short abbreviation (e.g., “NBT” for “Numbers and Operations in Base Ten”). Domains are larger groups of related standards. Each domain can further be broken down into clusters of related standards (e.g., “Generalize place value understanding for multi-digit whole numbers”). For high school, the organization of the standards has transitioned from the use of domain and cluster to ‘Ōpa‘a (unit or set) to reflect a more coherent storyline for high school by suggesting a sequencing of the standards.

3. **Standard**
   This element specifies the standard within a particular domain / ‘ōpa‘a.

This coding scheme is illustrated in the following figure:

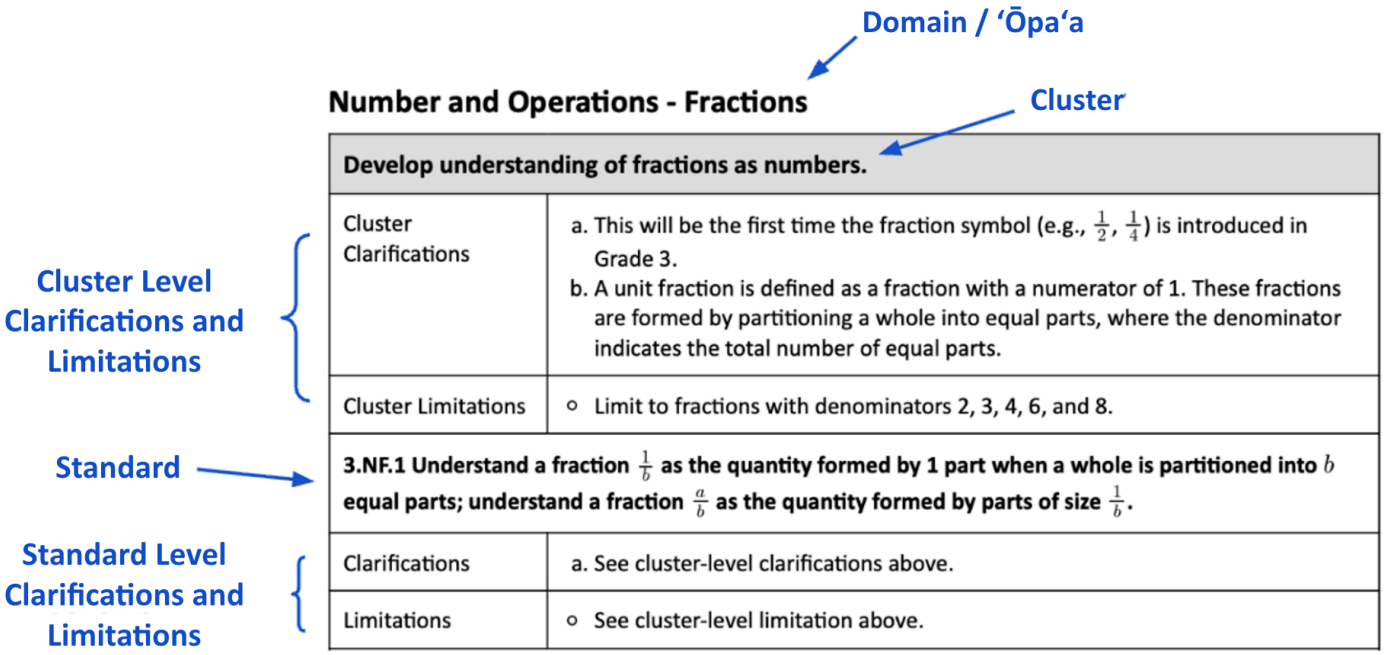


While **clusters** are no longer included in the coding scheme (because each standard is uniquely numbered within each domain / ‘ōpa‘a), they continue to exist as a way of further grouping tightly related standards within a domain.

Clusters and standards may be accompanied by clarifications and limitations. A clarification further explains/details expectations around a cluster or standard and can help specify what should be assessed as part of the standard. They must be viewed as supporting the standard rather than representing the standard in its entirety. A limitation describes the boundaries and restrictions in the scope of a particular cluster or standard. To easily distinguish between clarifications and limitations,

the former is lettered and the latter is bulleted. It should be noted that there are no cluster level clarifications and limitations for high school standards.

The following is an example from the Grade 3 standards. The domain is “Number and Operations - Fractions” and the cluster is “Develop understanding of fractions as numbers”. The clarifications and limitations at the cluster level apply to all standards within the cluster. The first standard in this cluster is 3.NF.1 where the “3” represents Grade 3, the “NF” represents the domain “Number and Operations - Fractions”, and the “1” represents the place in the sequence of standards within the domain.



Number and Operations - Fractions	
Develop understanding of fractions as numbers.	
Cluster Clarifications	a. This will be the first time the fraction symbol (e.g., $\frac{1}{2}$, $\frac{1}{4}$) is introduced in Grade 3. b. A unit fraction is defined as a fraction with a numerator of 1. These fractions are formed by partitioning a whole into equal parts, where the denominator indicates the total number of equal parts.
Cluster Limitations	○ Limit to fractions with denominators 2, 3, 4, 6, and 8.
3.NF.1 Understand a fraction $\frac{1}{b}$ as the quantity formed by 1 part when a whole is partitioned into b equal parts; understand a fraction $\frac{a}{b}$ as the quantity formed by parts of size $\frac{1}{b}$.	
Clarifications	a. See cluster-level clarifications above.
Limitations	○ See cluster-level limitation above.

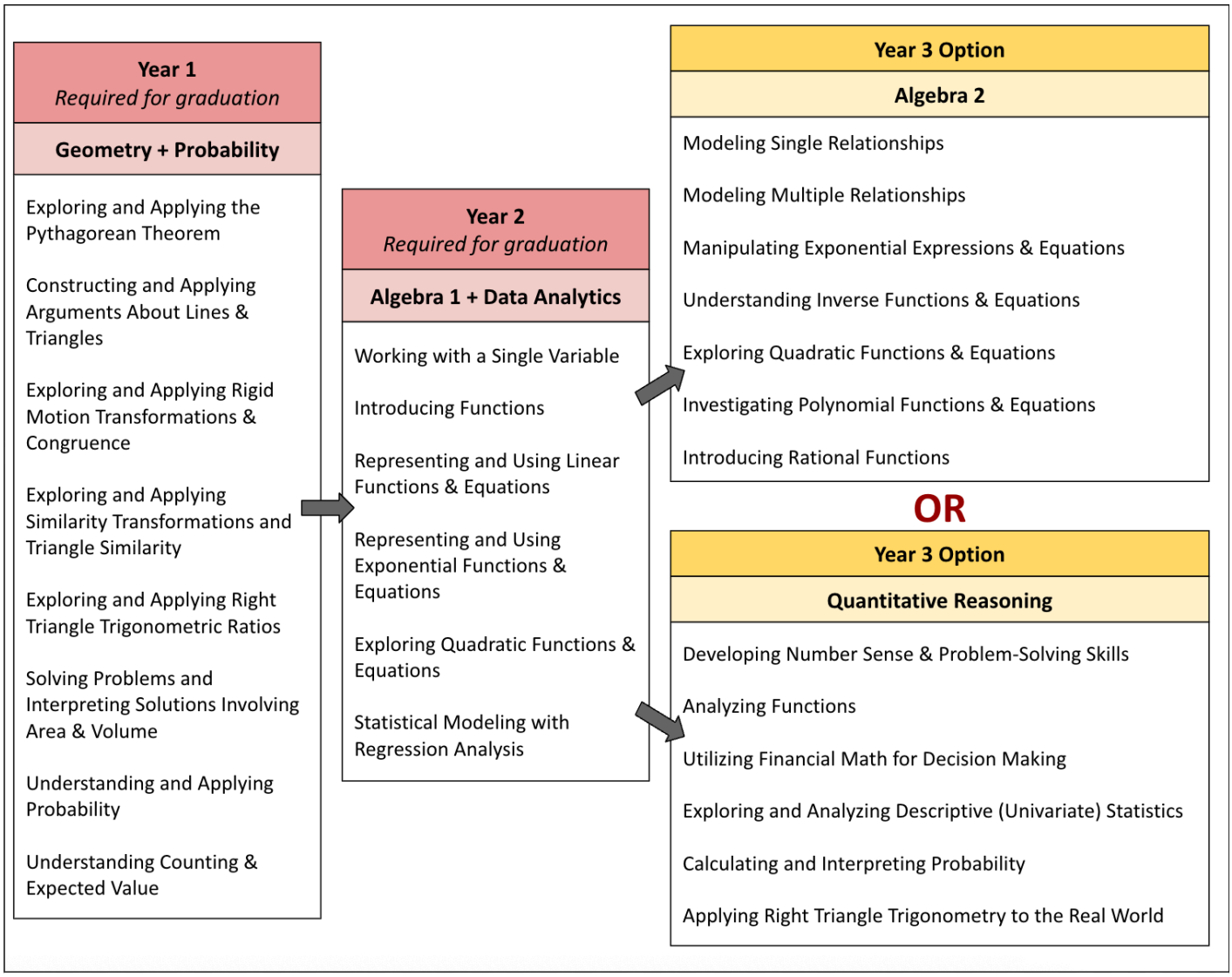
High School Overview

The standards for high school have been revised to clarify and streamline concepts while elevating the Standards for Mathematical Practice, resulting in a coherent order that carefully builds on knowledge introduced in previous grade levels. High school mathematics standards are organized by individual courses rather than as a single 9-12 “High School” grade band as was the case in the Common Core State Standards. Each cluster of standards is now called “ʻŌpaʻa” which is a grouping of related concepts.

The transition from 8th grade to 9th grade is crucial for students. Starting the high school mathematical experience with Geometry + Probability has several benefits including making clear real-world connections and fostering mathematical confidence. The Geometry + Probability standards have been revised to apply and reinforce proportional and linear algebraic concepts taught in middle school. By applying algebraic representations to geometric figures, students use variables as representations of real world quantities— angle measures, side lengths, areas. This provides a crucial mental anchor before they are asked to manipulate variables fluently beyond linear contexts. Algebra 1 + Data Analytics continues to explore linear, exponential and quadratic functions to model data and relationships.

In year three, students choose between two course options aligned to their post-secondary goals: Quantitative Reasoning or Algebra 2. Quantitative Reasoning is a new course designed for students who plan to pursue a major or a career which does not require calculus. Algebra 2 has been modified to build a stronger alignment and bridge to precalculus. This is illustrated in the figure on the next page.

High School Course ʻŌpaʻa At-a-Glance



Rationale for Geometry Before Algebra 1

A new sequence for high school mathematics is developed, where Geometry will be the default 9th grade mathematics course.

- The Geometry course will not require Algebra 1 as a prerequisite.
- Students who earn an Algebra 1 credit in the 8th grade enroll in Geometry in 9th grade. In the 10th grade, students have the option to take Quantitative Reasoning or Algebra 2.
- Students who did not take Algebra 1 in 8th grade (most students) enroll in Geometry in 9th grade, followed by Algebra 1 in 10th grade. In the 11th grade, students have the option to take Quantitative Reasoning or Algebra 2.

The proposed shift to a Geometry-first pathway in high school offers the following benefits:

- **Real-World Connections are More Obvious**: The applications of geometry are often more immediately apparent in the real world – architecture, design, art, navigation, etc. This can make the subject matter feel more relevant and motivating for 9th graders, potentially sparking greater interest in mathematics before diving into the more abstract manipulations of algebra.
- **Fosters Mathematical Confidence**: In high school, introducing geometry first allows students to develop a positive relationship with mathematics through its visual and hands-on nature, setting them up for greater success in algebra.
- **Supports Freshman Cohorts**: This proposed pathway enables schools to create 9th grade cohorts that include math as part of their teaming or academy structures. An additional benefit to having all students in Geometry is to have a greater diversity of perspectives when learning logic and reasoning. By changing this order, students on an accelerated track will still be able to take Algebra 1 in 8th grade, while there will be a common Geometry class for all 9th graders.
- **Alignment Leads to Smoother Transitions**: Students benefit from learning Geometry in 9th grade because the middle school geometry standards directly support the high school geometry standards. Moving Geometry to 9th grade also enables students to seamlessly transition from Algebra 1 directly to Quantitative Reasoning or Algebra 2 in the following year; schools that have made this change have noted that Algebra 2 students do not need as much "review" of Algebra 1 at the start of the year. Collectively, this allows students to have a more coherent mathematical experience.
- **Alignment with 9th Grade Sciences**: The concepts covered in middle school mathematics provide sufficient background knowledge needed for important applications commonly covered in 9th grade science courses. It is not essential for high school studies to begin with Algebra 1.

A Strategic Approach: Building a Stronger Foundation for Algebra 1 with Geometry First

The traditional Algebra 1 → Geometry sequence is well-established and valued for its focus on building procedural algebraic skills. However, many students struggle to move beyond procedure to true conceptual understanding, which can hinder their success in Algebra 1 and beyond. A Geometry-first sequence addresses this challenge head-on. It is not about delaying algebra, but about strategically front-loading the logical reasoning and contextual understanding that allow students to engage with Algebra 1 on a much deeper and more successful level.

1. **Addressing the "Abstraction Hurdle"**

   A common challenge is the abstract nature of algebraic concepts in Algebra 1, which can create early anxiety and a mindset that math is just about memorizing rules and contextually absent algorithms.

   - **A More Accessible On-Ramp**: Geometry leverages the concrete, visual-spatial skills they developed in 8th grade with transformations and the Pythagorean Theorem within real-world contexts. This allows them to experience success and build confidence at the start of their high school journey.
   - **Reinforcing Algebraic Thinking**: By applying algebra skills to geometric figures, students use variables to represent real world quantities —angle measures, side lengths, areas. This provides a crucial mental anchor before they are asked to manipulate variables fluently beyond linear contexts.

2. **Forging the "Why": How Geometry Emphasizes Problem Solving**

   Many teachers observe students who can follow the steps to solve an equation but falter when faced with a novel problem. This is often a gap in logical reasoning, not procedural skill. Geometry is the ideal environment to forge this skill.

   - **Building Logical Arguments**: The process of building a logical proof from a set of given information is a perfect analogue for solving a complex algebraic equation. Students learn to justify every step and understand how each one follows logically from the last. This is the very essence of algebraic reasoning.
   - **From "How" to "Why"**: After justifying claims in Geometry, a student is far better equipped to understand underlying algebraic concepts (e.g., why the distributive property works through the area model, or why they can perform the same operation on both sides of an equation). They move from mimicking a process to understanding its logical foundation.

3. **Algebra in Action: Reinforcing Skills Through Constant Application**

   A key concern is whether students will have the requisite algebra skills for Geometry. The reality is that a well-designed Geometry course is rich with algebraic application, providing constant, distributed practice.

   - **Contextualized Skill-Building**: Instead of practicing algebra in isolated problem sets, students are constantly using it as a tool to solve interesting geometric problems. They set up and solve equations to find missing angles in a transversal diagram or use the distance formula (a direct application of the Pythagorean Theorem) to prove properties of quadrilaterals.

- **Answering "When Will I Use This?"**: This integrated approach organically answers the ubiquitous question. Algebra becomes a necessary and powerful tool for discovery, not a set of abstract rules to be memorized. This contextual reinforcement can lead to better long-term retention of algebraic skills than an isolated Algebra 1 course.

By sequencing Geometry first, we give students the "why" before the "how." They enter Algebra 1 with stronger logical faculties, a better grasp of what variables can represent, and a year of experience applying algebraic tools in meaningful contexts. It is a proactive strategy to build more resilient, thoughtful, and ultimately more successful mathematical problem-solvers.

Kindergarten

Counting and Cardinality

Know number names and the count sequence.	
K.CC.1: Count to 100 by ones and by tens.	
Clarifications	None
Limitations	◦ When counting by tens, only expected to master counting by the decade (10, 20, 30, ...). ◦ Does not require recognition of numerals.
K.CC.2: Count forward and backward beginning from a given number within the known sequence (instead of having to begin at 1).	
Clarifications	None
Limitations	◦ The “known sequence” can be within 20. The intent is to build toward addition and subtraction in first grade. ◦ Does not require recognition of numerals.
K.CC.3: Write numbers from 0 to 20. Represent a number of objects with a written numeral 0-20 (with 0 representing a count of no objects).	
Clarifications	None
Limitations	None
Count to tell the number of objects.	
Cluster Limitations	◦ Count objects within 20.
K.CC.4: Understand the relationship between numbers and quantities; connect counting to cardinality.	
Clarifications	a. When counting objects, say the number names in the standard order, pairing each object with one and only one number name and each number name with one and only one object. b. Understand that the last number name tells the number of objects counted. The number of objects is the same regardless of their arrangement or the order in which they were counted. c. Understand that each successive number name refers to a quantity that is one larger.
Limitations	◦ See cluster-level limitation above.

K.CC.5: Count to answer "how many?" questions about as many as 20 things arranged in a line, a rectangular array, or a circle, or as many as 10 things in a scattered configuration; given a number from 1–20, count out that many objects.	
Clarifications	None
Limitations	◦ See cluster-level limitation above.
Compare Numbers	
Cluster Limitations	◦ Group sizes are at most 10 objects each. ◦ The words greater than, less than, or equal to should be used instead of symbols.
K.CC.6: Identify whether the number of objects in one group is greater than, less than, or equal to the number of objects in another group, e.g., by using matching and counting strategies.	
Clarifications	None
Limitations	◦ See cluster-level limitations above.
K.CC.7: Compare two numbers between 1 and 10 presented as written numerals.	
Clarifications	None
Limitations	◦ See cluster-level limitations above.

Operations and Algebraic Thinking

Understand addition as putting together and adding to, and understand subtraction as taking apart and taking from.	
Cluster Limitations	◦ Solve addition and subtraction problems within 10. ◦ Exposure to addition and subtraction equations is expected. Writing of equations is encouraged, but not required. Note that it is not until Grade 1 that "Understand the meaning of the equal sign" is an expectation.
K.OA.1: Represent addition and subtraction with objects, fingers, mental images, drawings, sounds (e.g., claps), acting out situations, verbal explanations, expressions, or equations.	
Clarifications	None
Limitations	◦ Drawings do not need to show details but should show the mathematics in the problem. ◦ See cluster-level limitations above.

K.OA.2: Solve addition and subtraction word problems, and add and subtract within 10, e.g., by using objects or drawings to represent the problem.	
Clarifications	None
Limitations	◦ See cluster-level limitations above.
K.OA.3: Decompose numbers less than or equal to 10 into pairs in more than one way, e.g., by using objects or drawings, and record each decomposition by a drawing or equation (e.g., 5 = 2 + 3 and 5 = 4 + 1).	
Clarifications	None
Limitations	◦ See cluster-level limitations above.
K.OA.4: For any number from 1 to 9, find the number that makes 10 when added to the given number, e.g., by using objects or drawings, and record the answer with a drawing or equation.	
Clarifications	None
Limitations	◦ See cluster-level limitations above.
K.OA.5: Fluently add and subtract within 5.	
Clarifications	a. This standard does not require timed tests.
Limitations	None

Numbers and Operations in Base Ten

Work with numbers 11–19 to gain foundations for place value.	
K.NBT.1: Compose and decompose numbers from 11 to 19 into ten ones and some further ones, e.g., by using objects or drawings, and record each composition or decomposition by a drawing or equation (e.g., 18 = 10 + 8); understand that these numbers are composed of ten ones and one, two, three, four, five, six, seven, eight, or nine ones.	
Clarifications	a. Focuses on making a group of “ten ones” and some further ones. The transition to Grade 1 is to make “one ten” from that group of “ten ones.” b. Exposure to equations is expected but mastery of equations is not required.
Limitations	None

Measurement and Data

Describe and compare measurable attributes.	
K.MD.1: Describe measurable attributes of objects, such as length or weight. Describe several measurable attributes of a single object.	
Clarifications	None
Limitations	None
K.MD.2: Directly compare two objects with a measurable attribute in common, to see which object has "more of"/"less of" the attribute, and describe the difference. For example, directly compare the heights of two children and describe one child as taller/shorter.	
Clarifications	None
Limitations	None
Classify objects and count the number of objects in each category.	
Cluster Limitations	◦ Category counts are less than or equal to ten.
K.MD.3: Classify objects into given categories; count the numbers of objects in each category and sort the categories by count.	
Clarifications	None
Limitations	◦ See cluster-level limitation above.
K.MD.4: Generate questions to investigate situations within the classroom. Collect or consider data that can naturally answer questions by sorting and counting.	
Clarifications	None
Limitations	◦ See cluster-level limitation above.

Geometry

Identify and describe shapes (squares, circles, ovals, triangles, rectangles, rhombuses, hexagons, cubes, cones, cylinders, and spheres).	
K.G.1: Describe objects in the environment using names of shapes, and describe the relative positions of these objects using terms, such as above, below, beside, in front of, behind, and next to.	
Clarifications	a. Identify the following two-dimensional shapes based on attributes: square, circle, triangle, rectangle, hexagon, oval, rhombus. b. Identify the following three-dimensional shapes based on attributes: cubes, cones, cylinders, spheres.
Limitations	None
K.G.2: Correctly name shapes regardless of their orientations or overall size.	
Clarifications	a. Identify the following two-dimensional shapes based on attributes: square, circle, triangle, rectangle, hexagon, oval, rhombus. b. Identify the following three-dimensional shapes based on attributes: cubes, cones, cylinders, spheres.
Limitations	None
K.G.3: Identify shapes as two-dimensional (lying in a plane, "flat") or three-dimensional ("solid").	
Clarifications	a. Name flat shapes as two-dimensional or solid shapes as three-dimensional. Example: a square is flat but a cube is solid.
Limitations	None
Analyze, compare, create, and compose shapes.	
K.G.4: Analyze and compare two-and three-dimensional shapes, in different sizes and orientations, using informal language to describe their similarities, differences, parts (e.g., number of sides and vertices/"corners") and other attributes (e.g., having sides of equal length).	
Clarifications	None
Limitations	None
K.G.5: Model shapes in the world by building shapes from components (e.g., sticks and clay balls) and drawing shapes.	
Clarifications	None
Limitations	None

K.G.6: Compose simple shapes to form larger shapes. For example, “Can you join these two triangles with full sides touching to make a rectangle?”	
Clarifications	None
Limitations	None

Grade 1

Operations and Algebraic Thinking

Represent and solve problems involving addition and subtraction.	
Cluster Limitations	◦ Problems should be within 20. ◦ Not expected to know nor use the term inverse.
1.OA.1: Use addition and subtraction within 20 to solve word problems involving situations of adding to, taking from, putting together, taking apart, and comparing, with unknowns in all positions, e.g., by using objects, drawings, and equations with a symbol for the unknown number to represent the problem.	
Clarifications	None
Limitations	◦ See cluster-level limitations above.
1.OA.2: Solve word problems that call for addition of three whole numbers whose sum is less than or equal to 20, e.g., by using objects, drawings, and equations with a symbol for the unknown number to represent the problem.	
Clarifications	None
Limitations	◦ See cluster-level limitations above.
Understand and apply properties of operations and the relationship between addition and subtraction.	
Cluster Limitations	◦ Problems should be within 20.
1.OA.3: Apply properties of operations as strategies to add and subtract. Examples: If 8 + 3 = 11 is known, then 3 + 8 = 11 is also known. (commutative property of addition.) To add 2 + 6 + 4, the second two numbers can be added to make a ten, so 2 + 6 + 4 = 2 + 10 = 12. (associative property of addition.) [Students need not use formal terms for these properties.]	
Clarifications	None
Limitations	◦ See cluster-level limitation above.

1.OA.4: Understand subtraction as an unknown-addend problem. For example, subtract 10 – 8 by finding the number that makes 10 when added to 8.	
Clarifications	None
Limitations	◦ See cluster-level limitation above.
Add and subtract within 20.	
Cluster Limitations	◦ Problems should be within 20.
1.OA.5: Relate counting to addition and subtraction (e.g., by counting on 2 to add 2).	
Clarifications	None
Limitations	◦ See cluster-level limitation above.
1.OA.6: Add and subtract within 20, demonstrating fluency for addition and subtraction within 10. Use strategies such as counting on; making ten (e.g., 8 + 6 = 8 + 2 + 4 = 10 + 4 = 14); decomposing a number leading to a ten (e.g., 13 – 4 = 13 – 3 – 1 = 10 – 1 = 9); using the relationship between addition and subtraction (e.g., knowing that 8 + 4 = 12, one knows 12 – 8 = 4); and creating equivalent but easier or known sums (e.g., adding 6 + 7 by creating the known equivalent 6 + 6 + 1 = 12 + 1 = 13).	
Clarifications	a. This standard does not require timed tests.
Limitations	None
Work with addition and subtraction equations.	
Cluster Limitations	◦ Problems should be within 20.
1.OA.7: Understand the meaning of the equal sign, and determine if equations involving addition and subtraction are true or false. For example, which of the following equations are true and which are false? 6 = 6, 7 = 8 – 1, 5 + 2 = 2 + 5, 4 + 1 = 5 + 2.	
Clarifications	None
Limitations	◦ See cluster-level limitation above.
1.OA.8: Determine the unknown whole number in an addition or subtraction equation relating three whole numbers. For example, determine the unknown number that makes the equation true in each of the equations 8 + □ = 11, 5 = □ – 3, 6 + 6 = □.	
Clarifications	None
Limitations	◦ See cluster-level limitation above.

Numbers and Operations in Base Ten

Extend the counting sequence.	
1.NBT.1: Count to 120, starting at any number less than 120. In this range, read and write numerals and represent a number of objects with a written numeral.	
Clarifications	None
Limitations	None
Understand place value.	
1.NBT.2: Understand that the two digits of a two-digit number represent amounts of tens and ones.	
Clarifications	Understand the following as special cases: a. 10 can be thought of as a bundle of ten ones — called a "ten." b. The numbers from 11 to 19 are composed of a ten and one, two, three, four, five, six, seven, eight, or nine ones. c. The numbers 10, 20, 30, 40, 50, 60, 70, 80, 90 refer to one, two, three, four, five, six, seven, eight, or nine tens (and 0 ones).
Limitations	None
1.NBT.3: Compare two two-digit numbers based on meanings of the tens and ones digits, recording the results of comparisons with the symbols >, =, and <.	
Clarifications	None
Limitations	None
Use place value understanding and properties of operations to add and subtract.	
1.NBT.4: Add within 100, including adding a two-digit number and a one-digit number, and adding a two-digit number and a multiple of 10, using concrete models or drawings and strategies based on place value, properties of operations, and/or the relationship between addition and subtraction; relate the strategy to a written method and explain the reasoning used. Understand that in adding two-digit numbers, one adds tens and tens, ones and ones; and sometimes it is necessary to compose a ten.	
Clarifications	None
Limitations	None
1.NBT.5: Given a two-digit number, mentally find 10 more or 10 less than the number, without having to count; explain the reasoning used.	
Clarifications	None
Limitations	None

1.NBT.6: Subtract multiples of 10 in the range 10-90 from multiples of 10 in the range 10-90 (positive or zero differences), using concrete models or drawings and strategies based on place value, properties of operations, and/or the relationship between addition and subtraction; relate the strategy to a written method and explain the reasoning used.	
Clarifications	None
Limitations	None

Measurement and Data

Measure lengths indirectly and by iterating length units.	
1.MD.1: Order three objects by length; compare the lengths of two objects indirectly by using a third object.	
Clarifications	None
Limitations	None
1.MD.2: Express the length of an object as a whole number of length units, by laying multiple copies of a shorter object (the length unit) end to end; understand that the length measurement of an object is the number of same-size length units that span it with no gaps or overlaps. Limit to contexts where the object being measured is spanned by a whole number of length units with no gaps or overlaps.	
Clarifications	a. Include use of standard units such as square tiles or centimeter tiles.
Limitations	◦ Limit to contexts where the object being measured is spanned by a whole number of length units with no gaps or overlaps.
Work with time and money.	
1.MD.3: Tell and write time in hours and half-hours using analog and digital clocks.	
Clarifications	a. Tell and write time to the hour and half hour in everyday settings, paying attention to a.m. and p.m.
Limitations	◦ Not required to know the term elapsed time.
1.MD.4: Identify the value, number of cents, of a penny, nickel, dime and quarter and compare the value of single coins.	
Clarifications	a. Only compare the single value of the coins.
Limitations	None

Represent and interpret data.	
1.MD.5: Organize, represent, and interpret data with up to three categories; ask and answer questions about the total number of data points, how many in each category, and how many more or less are in one category than in another.	
Clarifications	a. Use tally marks and numerical values to organize and represent data. b. Summarize the number of tally marks in each category.
Limitations	◦ Number of categories presented in graphs and charts should be within 3. ◦ Use single-unit scaled graphs. ◦ Number of data points should be within 20.

Geometry

Reason with shapes and their attributes.	
1.G.1: Distinguish between defining attributes (e.g., triangles are closed and three-sided) versus non-defining attributes (e.g., color, orientation, overall size); build and draw shapes to possess defining attributes.	
Clarifications	None
Limitations	None
1.G.2: Compose two-dimensional shapes (rectangles, squares, trapezoids, triangles, half-circles, and quarter-circles) or three-dimensional shapes (cubes, right rectangular prisms, right circular cones, and right circular cylinders) to create a composite shape, and compose new shapes from the composite shape.	
Clarifications	a. Use these common two-dimensional shapes to create composite shapes: circles, half-circles, quarter-circles, triangles, squares, rectangles (Students should know that a square is a type of rectangle, based on its attributes.), hexagons, trapezoids. b. Use these common three-dimensional shapes to create composite shapes: cubes, cones, cylinders, spheres, rectangular prisms, right circular cones, right circular cylinders.
Limitations	◦ The size of the shape does not necessarily distinguish between common and composite. ◦ Do not need to learn formal names, such as, “right rectangular prism”.

1.G.3: Partition circles and rectangles into two and four equal shares, describe the shares using the words halves, fourths, and quarters, and use the phrases half of, fourth of, and quarter of. Describe the whole as two of, or four of the shares. Understand for these examples that decomposing into more equal shares creates smaller shares.	
Clarifications	None
Limitations	◦ No shading of the shares is needed, only required to partition the whole shape into equal shares. ◦ Not expected to write the fraction using fraction notation.

Grade 2

Operations and Algebraic Thinking

Represent and solve problems involving addition and subtraction.	
2.OA.1: Use addition and subtraction within 100 to solve one- and two-step word problems involving situations of adding to, taking from, putting together, taking apart, and comparing, with unknowns in all positions, e.g., by using drawings and equations with a symbol for the unknown number to represent the problem.	
Clarifications	None
Limitations	◦ The sum of the numbers should be no greater than 1000. ◦ Two step problems should be limited to single-digit addends.
Add and subtract within 20.	
2.OA.2: Fluently add and subtract within 20 using mental strategies. By end of Grade 2, know from memory all sums of two one-digit numbers. [See standard 1.OA.6 for a list of mental strategies.]	
Clarifications	a. This standard does not require timed tests.
Limitations	None
Work with equal groups of objects to gain foundations for multiplication.	
2.OA.3: Determine whether a group of objects (up to 20) has an odd or even number of members, e.g., by pairing objects or counting them by 2s; write an equation to express an even number as a sum of two equal addends.	
Clarifications	None
Limitations	None
2.OA.4: Use addition to find the total number of objects arranged in rectangular arrays with up to 5 rows and up to 5 columns; write an equation to express the total as a sum of equal addends.	
Clarifications	None
Limitations	None

Numbers and Operations in Base Ten

Understand place value.	
2.NBT.1: Understand that the three digits of a three-digit number represent amounts of hundreds, tens, and ones; e.g., 706 equals 7 hundreds, 0 tens, and 6 ones.	
Clarifications	a. 100 can be thought of as a bundle of ten tens — called a “hundred.” b. The numbers 100, 200, 300, 400, 500, 600, 700, 800, 900 refer to one, two, three, four, five, six, seven, eight, or nine hundreds (and 0 tens and 0 ones).
Limitations	None
2.NBT.2: Count within 1000; skip-count by 2s, 5s, 10s, and 100s.	
Clarifications	None
Limitations	None
2.NBT.3: Read and write numbers to 1000 using base-ten numerals, number names, and expanded form.	
Clarifications	None
Limitations	None
2.NBT.4: Compare two three-digit numbers based on meanings of the hundreds, tens, and ones digits, using >, =, and < symbols to record the results of comparisons.	
Clarifications	None
Limitations	None
Use place value understanding and properties of operations to add and subtract.	
2.NBT.5: Fluently add and subtract within 100 using strategies based on place value, properties of operations, and/or the relationship between addition and subtraction.	
Clarifications	a. This standard does not require timed tests.
Limitations	◦ The sum of the numbers should be no greater than 100.

2.NBT.6: Add up to four two-digit numbers using strategies based on place value and properties of operations.	
Clarifications	None
Limitations	None
2.NBT.7: Add and subtract within 1000, using concrete models or drawings and strategies based on place value, properties of operations, and/or the relationship between addition and subtraction; relate the strategy to a written method. Understand that in adding or subtracting three-digit numbers, one adds or subtracts hundreds and hundreds, tens and tens, ones and ones; and sometimes it is necessary to compose or decompose tens or hundreds.	
Clarifications	None
Limitations	None
2.NBT.8: Mentally add 10 or 100 to a given number 100–900, and mentally subtract 10 or 100 from a given number 100–900.	
Clarifications	None
Limitations	None
2.NBT.9: Explain why addition and subtraction strategies work, using place value and the properties of operations.	
Clarifications	a. Explanations may be supported by drawings or objects.
Limitations	None

Measurement and Data

Measure and estimate lengths in standard units.	
2.MD.1: Measure the length of an object by selecting and using appropriate tools such as rulers, yardsticks, meter sticks, and measuring tapes.	
Clarifications	a. Use the customary measurement system of units (e.g., feet, inches, yards) and metric system of units (e.g., meters, centimeters, millimeters).
Limitations	None

2.MD.2: Measure the length of an object twice, using length units of different lengths for the two measurements; describe how the two measurements relate to the size of the unit chosen.	
Clarifications	None
Limitations	None
2.MD.3: Estimate lengths using units of inches, feet, centimeters, and meters.	
Clarifications	None
Limitations	None
2.MD.4: Measure to determine how much longer one object is than another, expressing the length difference in terms of a standard length unit.	
Clarifications	None
Limitations	None
Relate addition and subtraction to length.	
2.MD.5: Use addition and subtraction within 100 to solve word problems involving lengths that are given in the same units, e.g., by using drawings (such as drawings of rulers) and equations with a symbol for the unknown number to represent the problem.	
Clarifications	None
Limitations	None
2.MD.6: Represent whole numbers as lengths from 0 on a number line diagram with equally spaced points corresponding to the numbers 0, 1, 2, ..., and represent whole-number sums and differences within 100 on a number line diagram.	
Clarifications	None
Limitations	None
Work with time and money.	
2.MD.7: Tell and write time from analog and digital clocks to the nearest five minutes, using a.m. and p.m.	
Clarifications	None
Limitations	◦ Problems involving elapsed time should avoid crossing over a.m. and p.m.

2.MD.8: Solve word problems within a dollar (using coins) and within $100 (using dollar bills) including problems that require finding values of different combinations of coins or dollars bills, using $ and ¢ symbols appropriately.	
Clarifications	a. This is the first time students are required to find the value of a group of coins. b. The total quantity should be based on cents and the value of a group of coins should be less than 100 cents. c. Dollar bills may include $1, $5, $10, $20, and $100. d. The $ symbol should only be used when referring to whole dollar amounts.
Limitations	◦ Problems used should have only dollars or only cents. ◦ Not expected to write decimal numbers for coin amounts.
Represent and interpret data.	
2.MD.9: Collect or consider data that can naturally answer questions by using measurements with whole-number units. Create questions generated by measurement data by measuring lengths of several objects to the nearest whole unit, or by making repeated measurements of the same object. Show the measurements by making a line plot, where the horizontal scale is marked off in whole-number increments.	
Clarifications	None
Limitations	None
2.MD.10: Draw a picture graph and a bar graph (with single-unit scale) to represent a data set with up to four categories. Solve simple put-together, take-apart, and compare problems using information presented in a bar graph.	
Clarifications	None
Limitations	◦ Data sets should be no more than four categories.

Geometry

Reason with shapes and their attributes.	
2.G.1: Recognize and draw shapes having specified attributes, such as a given number of angles or a given number of equal faces. Identify triangles, quadrilaterals, pentagons, hexagons, and cubes.	
Clarifications	a. Attributes are compared directly or visually, not compared by measuring.
Limitations	None

2.G.2: Partition a rectangle into rows and columns of same-size squares and count to find the total number of them.	
Clarifications	None
Limitations	◦ Rectangles should be no larger than 5 rows and 5 columns.
2.G.3: Partition circles and rectangles into two, three, or four equal shares, describe the shares using the words halves, thirds, half of, a third of, etc., and describe the whole as two halves, three thirds, four fourths. Recognize that equal shares of identical wholes need not have the same shape.	
Clarifications	None
Limitations	◦ No shading of the shares is needed, only required to partition the whole shape into equal shares.

Grade 3

Operations and Algebraic Thinking

Represent and solve problems involving multiplication and division.	
Cluster Limitations	◦ Multiply and divide within 100. ◦ The quotient is a whole number.
3.OA.1: Interpret products of whole numbers, e.g., interpret 5 × 7 as the total number of objects in 5 groups of 7 objects each. For example, describe a context in which a total number of objects can be expressed as 5 × 7.	
Clarifications	a. This standard does not include calculating products. It is about understanding the meaning of each of the factors in 5 x 7, not calculating the product of 5 x 7.
Limitations	◦ See cluster-level limitations above.
3.OA.2: Interpret whole-number quotients of whole numbers, e.g., interpret 56 ÷ 8 as the number of objects in each share when 56 objects are partitioned equally into 8 shares, or as a number of shares when 56 objects are partitioned into equal shares of 8 objects each. For example, describe a context in which a number of shares or a number of groups can be expressed as 56 ÷ 8.	
Clarifications	a. This standard does not include calculating quotients. It is about understanding the meaning of 56 ÷ 8, not calculating the quotient of 56 ÷ 8.
Limitations	◦ See cluster-level limitations above.
3.OA.3: Use multiplication and division within 100 to solve word problems in situations involving equal groups, arrays, and measurement quantities, e.g., by using drawings and equations with a symbol for the unknown number to represent the problem.	
Clarifications	a. Equation is used instead of “number sentence,” allowing the same term to be used throughout K-12. b. The common multiplication and division situations for Grade 3 are equal groups, arrays, area, and measurement quantities.
Limitations	◦ See cluster-level limitations above.

3.OA.4: Determine the unknown whole number in a multiplication or division equation relating three whole numbers. For example, determine the unknown number that makes the equation true in each of the equations 8 × ? = 48, 5 = ? ÷ 3, 6 × 6 = ?.	
Clarifications	a. The unknown can be in any position in the equation.
Limitations	◦ See cluster-level limitations above.
Understand properties of multiplication and the relationship between multiplication and division.	
Cluster Clarifications	◦ Use the properties of operations to solve multiplication and division problems.
Cluster Limitations	◦ Multiply and divide within 100. ◦ Not expected to formally name or identify the properties (e.g., Commutative property of multiplication, Associative property of multiplication, and Distributive property) ◦ Not expected to memorize multiplication and division facts.
3.OA.5: Apply properties of operations as strategies to multiply and divide. Examples: If 6 × 4 = 24 is known, then 4 × 6 = 24 is also known (Commutative property of multiplication). 3 × 5 × 2 can be found by 3 × 5 = 15, then 15 × 2 = 30, or by 5 × 2 = 10, then 3 × 10 = 30 (Associative property of multiplication). Knowing that 8 × 5 = 40 and 8 × 2 = 16, one can find 8 × 7 as 8 × (5 + 2) = (8 × 5) + (8 × 2) = 40 + 16 = 56 (Distributive property).	
Clarifications	a. See cluster-level clarification above.
Limitations	◦ See cluster-level limitations above.
3.OA.6: Understand division as an unknown-factor problem. For example, find 32 ÷ 8 by finding the number that makes 32 when multiplied by 8.	
Clarifications	a. This standard does not require computation but understanding the conceptual relationship between multiplication and division.
Limitations	◦ See cluster-level limitations above.
Multiply and divide within 100.	
3.OA.7: Fluently multiply and divide within 100, using strategies such as the relationship between multiplication and division (e.g., knowing that 8 × 5 = 40, one knows 40 ÷ 5 = 8) or properties of operations. By the end of Grade 3, know from memory all products of two one-digit numbers.	
Clarifications	a. This standard does not require timed tests. b. Not expected to formally name or identify the properties (e.g., Commutative property of multiplication, Associative property of multiplication, and Distributive property). c. This includes two-digit × one-digit when the answer is less than or equal to 100.
Limitations	None

Solve problems involving the four operations, and identify and explain patterns in arithmetic.	
3.OA.8: Solve two-step word problems using the four operations with whole numbers and having whole-number answers. Represent these problems using equations with a letter standing for the unknown quantity and assess the reasonableness of answers using mental math and estimation.	
Clarifications	None
Limitations	◦ Products and quotients must stay within 100.
3.OA.9: Identify arithmetic patterns (including patterns in the addition table or multiplication table), and explain them using properties of operations. For example, observe that 4 times a number is always even, and explain why 4 times a number can be decomposed into two equal addends.	
Clarifications	None
Limitations	◦ Multiplication and division problems should be within 100. ◦ Addition and subtraction steps should involve numbers larger than 100 with solutions within 1000.

Number and Operations in Base Ten

Use place value understanding and properties of operations to perform multi-digit arithmetic.	
3.NBT.1: Use place value understanding to round whole numbers to the nearest 10 or 100.	
Clarifications	a. Building understanding of rounding by working with numbers within 100 first and then progressing to 1000. b. This standard intends that students round numbers in ways that are useful for problem solving. Therefore, rounding three-digit numbers to hundreds and two-digit numbers to tens should be emphasized.
Limitations	◦ Numbers are limited to less than 1000.
3.NBT.2: Fluently add and subtract within 1000 using strategies and algorithms based on place value, properties of operations, and/or the relationship between addition and subtraction.	
Clarifications	a. This standard does not require timed tests. b. Not expected to formally name or identify the properties (e.g., Commutative property of multiplication, Associative property of multiplication, and Distributive property).
Limitations	None

3.NBT.3: Multiply one-digit whole numbers by multiples of 10 in the range 10-90 (e.g., 9 × 80, 5 × 60) using strategies based on place value and properties of operations.	
Clarifications	a. Understand that as numbers are multiplied by a multiple of 10, the digit shifts left, making the value ten times more with each shift. b. Not expected to formally name or identify the properties (e.g., Commutative property of multiplication, Associative property of multiplication, and Distributive property). c. Not expected to use the standard algorithm.
Limitations	None

Number and Operations - Fractions

Develop understanding of fractions as numbers.	
Cluster Clarifications	◦ This will be the first time the fraction symbol (e.g., $\frac{1}{2}$, $\frac{1}{4}$) is introduced in Grade 3. ◦ A unit fraction is defined as a fraction with a numerator of 1. These fractions are formed by partitioning a whole into equal parts, where the denominator indicates the total number of equal parts.
Cluster Limitations	◦ Limit to fractions with denominators 2, 3, 4, 6, and 8.
3.NF.1: Understand a fraction $\frac{1}{b}$ as the quantity formed by 1 part when a whole is partitioned into b equal parts; understand a fraction $\frac{a}{b}$ as the quantity formed by a parts of size $\frac{1}{b}$.	
Clarifications	a. See cluster-level clarifications above.
Limitations	◦ See cluster-level limitation above.
3.NF.2: Understand a fraction as a number on the number line; represent fractions on a number line diagram.	
Clarifications	a. Represent a fraction $\frac{1}{b}$ on a number line diagram by defining the interval from 0 to 1 as the whole and partitioning it into b equal parts. Recognize that each part has size $\frac{1}{b}$ and that the endpoint of the part based at 0 locates the number $\frac{1}{b}$ on the number line. b. Represent a fraction $\frac{a}{b}$ on a number line diagram by marking off a lengths $\frac{1}{b}$ from 0. Recognize that the resulting interval has size $\frac{a}{b}$ and that its endpoint locates the number $\frac{a}{b}$ on the number line. c. See cluster-level clarifications above.
Limitations	◦ See cluster-level limitation above.

3.NF.3: Explain equivalence of fractions in special cases, and compare fractions by reasoning about their size.	
Clarifications	a. Understand two fractions as equivalent (equal) if they are the same size, or the same point on a number line. b. Recognize and generate simple equivalent fractions, e.g., $\frac{1}{2} = \frac{2}{4}$, $\frac{4}{6} = \frac{2}{3}$). Explain why the fractions are equivalent, e.g., by using a visual fraction model, pattern blocks, Cuisenaire rods. c. Express whole numbers as fractions, and recognize fractions that are equivalent to whole numbers. Examples: Express 3 in the form $3 = \frac{3}{1}$; recognize that $\frac{6}{1} = 6$; locate $\frac{4}{4}$ and 1 at the same point of a number line diagram. d. Compare two fractions with the same numerator or the same denominator by reasoning about their size. Recognize that comparisons are valid only when the two fractions refer to the same whole. Record the results of comparisons with the symbols >, =, or <, and justify the conclusions, e.g., by using a visual fraction model, pattern blocks, Cuisenaire rods. e. See cluster-level clarifications above.
Limitations	◦ See cluster-level limitation above.

Measurement and Data

Solve problems involving measurement and estimation of intervals of time, liquid volumes, and masses of objects.	
3.MD.1: Tell and write time to the nearest minute and measure time intervals in minutes. Solve word problems involving addition and subtraction of time intervals in minutes, e.g., by representing the problem on a number line diagram.	
Clarifications	a. This includes one-step problems that cross into a new hour.
Limitations	◦ Does not require a.m. and p.m. understanding beyond exposure.
3.MD.2: Measure and estimate the mass and liquid volume of objects using grams (g), kilograms (kg), and liters (L). Solve one-step word problems involving these measurements using addition, subtraction, multiplication, or division, where all measurements are in the same unit (e.g., grams, kilograms, or liters).	
Clarifications	a. This includes one-step word problems using the measures in the same units (no conversion of units).
Limitations	◦ Excludes compound units such as cm^3 and finding the geometric volume of a container.

Represent and interpret data	
3.MD.3: Draw a scaled picture graph and a scaled bar graph to represent a data set with several categories. Solve one- and two-step "how many more" and "how many less" problems using information presented in scaled bar graphs. For example, draw a bar graph in which each square in the bar graph might represent 5 pets.	
Clarifications	a. The scale is not always 1:1.
Limitations	◦ Limit to a scaled picture graph or a scaled bar graph.
3.MD.4: Generate measurement data by measuring lengths using rulers marked with halves and fourths of an inch. Show the data by making a line plot, where the horizontal scale is marked off in appropriate units— whole numbers, halves, or quarters.	
Clarifications	None
Limitations	◦ Limit to line plot using whole numbers, halves, and fourths (quarters) of an inch for measuring lengths using rulers.
Geometric measurement: Understand concepts of area and relate area to multiplication and to addition.	
3.MD.5: Recognize area as an attribute of plane figures and understand concepts of area measurement.	
Clarifications	a. A square with side length 1 unit, called "a unit square," is said to have "one square unit" of area, and can be used to measure area. b. A plane figure which can be covered without gaps or overlaps by n unit squares is said to have an area of n square units.
Limitations	◦ Square units (no specific measurement system yet)
3.MD.6: Measure areas by counting unit squares (square cm, square m, square in, square ft, and improvised units).	
Clarifications	a. Improvised units (also called non-standard units) are any consistent, same-sized units that are not formal measurement tools, but are used to measure area.
Limitations	None

3.MD.7: Relate area to the operations of multiplication and addition.	
Clarifications	a. Find the area of a rectangle with whole-number side lengths by tiling it, and show that the area is the same as would be found by multiplying the side lengths. b. Multiply side lengths to find areas of rectangles with whole-number side lengths in the context of solving real world and mathematical problems, and represent whole-number products as rectangular areas in mathematical reasoning. c. Use tiling to show in a concrete case that the area of a rectangle with whole-number side lengths a and $b + c$ is the sum of $a \times b$ and $a \times c$. Use area models to represent the distributive property in mathematical reasoning. d. Recognize area as additive. Find areas of rectilinear figures by decomposing them into non-overlapping rectangles and adding the areas of the non-overlapping parts, applying this technique to solve real-world problems. e. Not expected to formally name or identify the Distributive property.
Limitations	None
Geometric measurement: Recognize perimeter as an attribute of plane figures and distinguish between linear and area measures.	
3.MD.8: Solve real world and mathematical problems involving perimeters of polygons, including finding the perimeter given the side lengths, finding an unknown side length, and exhibiting rectangles with the same perimeter and different areas or with the same area and different perimeters.	
Clarifications	a. Use shapes such as rectangles, squares, and other polygons with whole number side lengths.
Limitations	None

Geometry

Reason with shapes and their attributes	
3.G.1: Understand that shapes in different categories (e.g., rhombuses, rectangles, and others) may share attributes (e.g., having four sides), and that the shared attributes can define a larger category (e.g., quadrilaterals). Recognize rhombuses, rectangles, and squares as examples of quadrilaterals, and draw examples of quadrilaterals that do not belong to any of these subcategories.	
Clarifications	a. Using common attributes such as number of sides, number of angles, and side lengths (equal or unequal). b. Classify by visual attributes only and not based on calculations (such as measuring side lengths or angles).
Limitations	None

3.G.2: Partition shapes into parts with equal areas. Express the area of each part as a unit fraction of the whole. For example, partition a shape into 4 parts with equal area, and describe the area of each part as $\frac{1}{4}$ of the area of the shape.	
Clarifications	a. Emphasis on understanding (not calculating) that each part of the shape represents an equal fraction of the total area.
Limitations	◦ Limit to 2, 3, 4, 6 or 8 equal parts. ◦ Limit unit fractions to $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{6}$, and $\frac{1}{8}$. ◦ Limit to area and equal partitioning.

Grade 4

Operations and Algebraic Thinking

Use the four operations with whole numbers to solve problems.	
Cluster Limitations	◦ Solutions within 100.
4.OA.1: Interpret a multiplication equation as a comparison, e.g., interpret 35 = 5 × 7 as a statement that 35 is 5 times as many as 7 and 7 times as many as 5. Represent verbal statements of multiplicative comparisons as multiplication equations.	
Clarifications	None
Limitations	◦ See cluster-level limitation above.
4.OA.2: Multiply or divide to solve word problems involving multiplicative comparison, e.g., by using drawings and equations with a symbol for the unknown number to represent the problem, distinguishing multiplicative comparison from additive comparison.	
Clarifications	None
Limitations	◦ See cluster-level limitation above.
4.OA.3: Solve multistep word problems posed with whole numbers and having whole-number answers using the four operations, including problems in which remainders must be interpreted. Represent these problems using equations with a letter standing for the unknown quantity. Assess the reasonableness of answers using mental computation and estimation strategies, including rounding.	
Clarifications	None
Limitations	◦ See cluster-level limitation above.
Gain familiarity with factors and multiples.	
4.OA.4: Find all factor pairs for a whole number in the range 1–100. Recognize that a whole number is a multiple of each of its factors. Determine whether a given whole number in the range 1–100 is a multiple of a given one-digit number. Determine whether a given whole number in the range 1–100 is prime or composite.	
Clarifications	None
Limitations	◦ When finding factor pairs, use any factor pair with a product within 100.

Generate and analyze patterns.	
4.OA.5: Generate a sequence of numbers or shape pattern that follows a specific rule. Identify patterns within the sequence that are not directly stated in the original rule.	
Clarifications	a. Original rule should be given.
Limitations	None

Number and Operations in Base Ten

Generalize place value understanding for multi-digit whole numbers.	
Cluster Limitations	◦ Limited to whole numbers less than or equal to 1,000,000.
4.NBT.1: Recognize that in a multi-digit whole number, a digit in one place represents ten times what it represents in the place to its right. For example, recognize that 700 ÷ 70 = 10 by applying concepts of place value and division.	
Clarifications	a. In the number 700, the digit 7 is in the hundreds place. In the number 70, the digit 7 is in the tens place. The digit 7 in the hundreds place is 10 times greater than the digit 7 in the tens place. Therefore, 700 ÷ 70 = 10.
Limitations	◦ See cluster-level limitation above.
4.NBT.2: Read and write multi-digit whole numbers using base-ten numerals, number names, and expanded form. Compare two multi-digit numbers based on meanings of the digits in each place, using >, =, and < symbols to record the results of comparisons.	
Clarifications	None
Limitations	◦ See cluster-level limitation above.
4.NBT.3: Use place value understanding to round multi-digit whole numbers to any place.	
Clarifications	None
Limitations	◦ See cluster-level limitation above.

Use place value understanding and properties of operations to perform multi-digit arithmetic.	
4.NBT.4: Fluently add and subtract multi-digit whole numbers using the standard algorithm.	
Clarifications	a. This standard does not require timed tests.
Limitations	None
4.NBT.5: Multiply a whole number of up to four digits by a one-digit whole number, and multiply two two-digit numbers, using strategies based on place value and the properties of operations. Illustrate and explain the calculation by using equations, rectangular arrays, and/or area models.	
Clarifications	a. Not expected to use standard algorithm for multiplication.
Limitations	None
4.NBT.6: Find whole-number quotients and remainders with up to four-digit dividends and one-digit divisors, using strategies based on place value, the properties of operations, and/or the relationship between multiplication and division. Illustrate and explain the calculation by using equations, rectangular arrays, and/or area models.	
Clarifications	a. Not expected to use standard algorithm for division.
Limitations	None

Number and Operations - Fractions

Extend understanding of fraction equivalence and ordering.	
Cluster Clarifications	◦ Fractions do not need to be in simplest form.
Cluster Limitations	◦ Limit to fractions with denominators 2, 3, 4, 5, 6, 8, 10, 12, and 100.
4.NF.1: Explain why a fraction $\frac{a}{b}$ is equivalent to a fraction $\frac{(n \times a)}{(n \times b)}$ by using visual fraction models, with attention to how the number and size of the parts differ even though the two fractions themselves are the same size. Use this principle to recognize and generate equivalent fractions.	
Clarifications	a. Multiplying the numerator and denominator by the same number produces an equivalent fraction. b. See cluster-level clarification above.
Limitations	◦ See cluster-level limitation above.

4.NF.2: Compare two fractions with different numerators and different denominators, e.g., by creating common denominators or numerators, or by comparing to a benchmark fraction such as $\frac{1}{2}$. Recognize that comparisons are valid only when the two fractions refer to the same whole. Record the results of comparisons with symbols >, =, or <, and justify the conclusions, e.g., by using a visual fraction model.	
Clarifications	a. See cluster-level clarification above.
Limitations	◦ See cluster-level limitation above.
Build fractions from unit fractions by applying and extending previous understandings of operations on whole numbers.	
4.NF.3: Understand a fraction $\frac{a}{b}$ with a > 1 as a sum of fractions $\frac{1}{b}$.	
Clarifications	a. Understand addition and subtraction of fractions as joining and separating parts referring to the same whole. b. Decompose a fraction into a sum of fractions with the same denominator in more than one way, recording each decomposition by an equation. Justify decompositions, e.g., by using a visual fraction model. Examples: $\frac{3}{8} = \frac{1}{8} + \frac{1}{8} + \frac{1}{8}$; $\frac{3}{8} = \frac{1}{8} + \frac{2}{8}$; $2\frac{1}{8} = 1 + 1 + \frac{1}{8} = \frac{8}{8} + \frac{8}{8} + \frac{1}{8}$. c. Add and subtract mixed numbers with like denominators, e.g., by replacing each mixed number with an equivalent fraction, and/or by using properties of operations and the relationship between addition and subtraction. d. Solve word problems involving addition and subtraction of fractions referring to the same whole and having like denominators, e.g., by using visual fraction models and equations to represent the problem.
Limitations	None
4.NF.4: Apply and extend previous understandings of multiplication to multiply a fraction by a whole number.	
Clarifications	a. Understand a fraction $\frac{a}{b}$ as a multiple of $\frac{1}{b}$. For example, use a visual fraction model to represent $\frac{5}{4}$ as the product $5 \times (\frac{1}{4})$, recording the conclusion by the equation $\frac{5}{4} = 5 \times (\frac{1}{4})$. b. Understand a multiple of $\frac{a}{b}$ as a multiple of $\frac{1}{b}$, and use this understanding to multiply a fraction by a whole number. For example, use a visual fraction model to express $3 \times (\frac{2}{5})$ as $6 \times (\frac{1}{5})$, recognizing this product as $\frac{6}{5}$. (In general, $n \times (\frac{a}{b}) = \frac{(n \times a)}{b}$). c. Solve word problems involving multiplication of a fraction by a whole number, e.g., by using visual fraction models and equations to represent the problem. For example, if each person at a party will eat $\frac{3}{8}$ of a pound of roast beef, and there will be 5 people at the party, how many pounds of roast beef will be needed? Between what two whole numbers does your answer lie?
Limitations	None

Understand decimal notation for fractions, and compare decimal fractions.	
4.NF.5: Express a fraction with denominator 10 as an equivalent fraction with denominator 100, and use this technique to add two fractions with respective denominators 10 and 100. For example, express $\frac{3}{10}$ as $\frac{30}{100}$, and add $\frac{3}{10} + \frac{4}{100} = \frac{34}{100}$.	
Clarifications	None
Limitations	None
4.NF.6: Use decimal notation for fractions with denominators 10 or 100. For example, rewrite 0.62 as $\frac{62}{100}$; describe a length as 0.62 meters; locate 0.62 on a number line diagram.	
Clarifications	None
Limitations	None
4.NF.7: Compare two decimals to hundredths by reasoning about their size. Recognize that comparisons are valid only when the two decimals refer to the same whole. Record the results of comparisons with the symbols >, =, or <, and justify the conclusions, e.g., by using a visual model.	
Clarifications	None
Limitations	None

Measurement and Data

Solve problems involving measurement and conversion of measurements from a larger unit to a smaller unit.	
4.MD.1: Know relative sizes of measurement units within one system of units including km, m, cm; kg, g; lb, oz; l, ml; hr, min, sec. Within a single system of measurement, express measurements in a larger unit in terms of a smaller unit. Record measurement equivalents in a two-column table. For example, know that 1 ft is 12 times as long as 1 in. Express the length of a 4 ft snake as 48 in. Generate a conversion table for feet and inches listing the number pairs (1, 12), (2, 24), (3, 36), ...	
Clarifications	None
Limitations	None

4.MD.2: Use the four operations to solve word problems involving distances, intervals of time, liquid volumes, masses of objects, and money, including problems involving simple fractions, and problems that require expressing measurements given in a larger unit in terms of a smaller unit. Represent measurement quantities using diagrams such as number line diagrams that feature a measurement scale.	
Clarifications	None
Limitations	None
4.MD.3: Apply the area and perimeter formulas for rectangles in real world and mathematical problems. For example, find the width of a rectangular room given the area of the flooring and the length, by viewing the area formula as a multiplication equation with an unknown factor.	
Clarifications	None
Limitations	None
Represent and interpret data.	
4.MD.4: Make a line plot to display a data set of measurements in fractions of a unit ($\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$). Solve problems involving addition and subtraction of fractions by using information presented in line plots. For example, from a line plot find and interpret the difference in length between the longest and shortest specimens in an insect collection.	
Clarifications	None
Limitations	None
Geometric measurement: understand concepts of angle and measure angles.	
4.MD.5: Recognize angles as geometric shapes that are formed wherever two rays share a common endpoint, and understand concepts of angle measurement.	
Clarifications	a. An angle is measured with reference to a circle with its center at the common endpoint of the rays, by considering the fraction of the circular arc between the points where the two rays intersect the circle. An angle that turns through $\frac{1}{360}$ of a circle is called a "one-degree angle," and can be used to measure angles. b. An angle that turns through n one-degree angles is said to have an angle measure of n degrees.
Limitations	None
4.MD.6: Measure angles in whole-number degrees using a protractor. Sketch angles of specified measure.	
Clarifications	None
Limitations	None

4.MD.7: Recognize angle measure as additive. When an angle is decomposed into non-overlapping parts, the angle measure of the whole is the sum of the angle measures of the parts. Solve addition and subtraction problems to find unknown angles on a diagram in real world and mathematical problems, e.g., by using an equation with a symbol for the unknown angle measure.	
Clarifications	None
Limitations	None

Geometry

Draw and identify lines and angles, and classify shapes by properties of their lines and angles.	
4.G.1: Draw points, lines, line segments, rays, angles (right, acute, obtuse), and perpendicular and parallel lines. Identify these in two-dimensional figures.	
Clarifications	None
Limitations	None
4.G.2: Classify two-dimensional figures based on the presence or absence of parallel or perpendicular lines, or the presence or absence of angles of a specified size. Recognize right triangles as a category, and identify right triangles.	
Clarifications	None
Limitations	None
4.G.3: Recognize a line of symmetry for a two-dimensional figure as a line across the figure such that the figure can be folded along the line into matching parts. Identify line-symmetric figures and draw lines of symmetry.	
Clarifications	None
Limitations	None

Grade 5

Operations and Algebraic Thinking

Write and interpret numerical operations	
Cluster Limitations	◦ Exponents and nested grouping symbols are not included. ◦ Expressions do not include variables.
5.OA.1: Use parentheses, brackets, or braces in numerical expressions, and evaluate expressions with these symbols.	
Clarifications	a. Formal instruction on the full order of operations is not expected. b. Numerical expressions are allowed to include whole numbers, fractions, and decimals.
Limitations	◦ See cluster-level limitations above.
5.OA.2: Write simple expressions that record calculations with numbers, and interpret numerical expressions without evaluating them. For example, express the calculation "add 8 and 7, then multiply by 2" as 2 × (8 + 7). Recognize that 3 × (18932 + 921) is three times as large as 18932 + 921, without having to calculate the indicated sum or product.	
Clarifications	a. Understand the meaning of expressions without doing the calculation.
Limitations	◦ See cluster-level limitations above.
Analyze patterns and relationships.	
5.OA.3: Generate two numerical patterns using two given rules. Identify apparent relationships between corresponding terms. Form ordered pairs consisting of corresponding terms from the two patterns, and graph the ordered pairs on a coordinate plane. For example, given the rule "Add 3" and the starting number 0, and given the rule "Add 6" and the starting number 0, generate terms in the resulting sequences, and observe that the terms in one sequence are twice the corresponding terms in the other sequence. Explain informally why this is so.	
Clarifications	a. Graph ordered pairs only in Quadrant I of the coordinate plane, where both the x- and y-values are positive.
Limitations	None

Number and Operations in Base Ten

Understand the place value system	
5.NBT.1: Recognize that in a multi-digit number, a digit in one place represents 10 times as much as it represents in the place to its right and $\frac{1}{10}$ of what it represents in the place to its left.	
Clarifications	a. Not expected to compute or use exponents to represent powers of 10.
Limitations	◦ Decimals limited to tenths, hundredths, and thousandths.
5.NBT.2: Explain patterns in the number of zeros of the product when multiplying a number by powers of 10, and explain patterns in the placement of the decimal point when a decimal is multiplied or divided by a power of 10. Use whole-number exponents to denote powers of 10.	
Clarifications	a. The focus is on understanding and explaining patterns, not performing long or complex computations.
Limitations	None
5.NBT.3: Read, write, and compare decimals to thousandths.	
Clarifications	a. Read and write decimals to thousandths using base-ten numerals, number names, and expanded form, e.g., $347.392 = 3 \times 100 + 4 \times 10 + 7 \times 1 + 3 \times (\frac{1}{10}) + 9 \times (\frac{1}{100}) + 2 \times (\frac{1}{1000})$. b. Compare two decimals to thousandths based on meanings of the digits in each place, using >, =, and < symbols to record the results of comparisons.
Limitations	None
5.NBT.4: Use place value understanding to round decimals to any place.	
Clarifications	a. Provide context and use real-world situations like taxes, price of gas, tip on a bill, etc.
Limitations	◦ Limited to decimals up to the thousandths.
Perform operations with multi-digit whole numbers and with decimals to hundredths.	
5.NBT.5: Fluently multiply multi-digit whole numbers using the standard algorithm.	
Clarifications	a. This standard does not require timed tests.
Limitations	◦ Limited to 4-digit x 2-digit multiplication.

5.NBT.6: Find whole-number quotients of whole numbers with up to four-digit dividends and two-digit divisors, using strategies based on place value, the properties of operations, and/or the relationship between multiplication and division. Illustrate and explain the calculation by using equations, rectangular arrays, and/or area models.	
Clarifications	a. Conceptual understanding is the priority.
Limitations	◦ Should not use standard algorithm because it is a 6th grade standard.
5.NBT.7: Add, subtract, multiply, and divide decimals to hundredths, using concrete models or drawings and strategies based on place value, properties of operations, and/or the relationship between addition and subtraction; relate the strategy to a written method and explain the reasoning used.	
Clarifications	a. Conceptual understanding is the priority.
Limitations	◦ Should not use standard algorithm because it is a 6th grade standard.

Number and Operations - Fractions

Use equivalent fractions as a strategy to add and subtract fractions.	
5.NF.1: Add and subtract fractions with unlike denominators (including mixed numbers) by replacing given fractions with equivalent fractions in such a way as to produce an equivalent sum or difference of fractions with like denominators. For example, $\frac{2}{3} + \frac{5}{4} = \frac{8}{12} + \frac{15}{12} = \frac{23}{12}$. (In general, $\frac{a}{b} + \frac{c}{d} = \frac{(ad+bc)}{bd}$.)	
Clarifications	None
Limitations	None
5.NF.2: Solve word problems involving addition and subtraction of fractions referring to the same whole, including cases of unlike denominators, e.g., by using visual fraction models or equations to represent the problem. Use benchmark fractions and number sense of fractions to estimate mentally and assess the reasonableness of answers. For example, recognize an incorrect result $\frac{2}{5} + \frac{1}{2} = \frac{3}{7}$, by observing that $\frac{3}{7} < \frac{1}{2}$.	
Clarifications	None
Limitations	None

Apply and extend previous understandings of multiplication and division to multiply and divide fractions.	
5.NF.3: Interpret a fraction as division of the numerator by the denominator $(\frac{a}{b} = a \div b)$. Solve word problems involving division of whole numbers leading to answers in the form of fractions or mixed numbers, e.g., by using visual fraction models or equations to represent the problem. For example, interpret $\frac{3}{4}$ as the result of dividing 3 by 4, noting that $\frac{3}{4}$ multiplied by 4 equals 3, and that when 3 wholes are shared equally among 4 people each person has a share of size $\frac{3}{4}$. If 9 people want to share a 50-pound sack of rice equally by weight, how many pounds of rice should each person get? Between what two whole numbers does your answer lie?	
Clarifications	None
Limitations	None
5.NF.4: Apply and extend previous understandings of multiplication to multiply a fraction or whole number by a fraction.	
Clarifications	a. Interpret the product $(\frac{a}{b}) \times q$ as a parts of a partition of q into b equal parts; equivalently, as the result of a sequence of operations $a \times q \div b$. For example, use a visual fraction model to show $(\frac{2}{3}) \times 4 = \frac{8}{3}$, and create a story context for this equation. Do the same with $(\frac{2}{3}) \times (\frac{4}{5}) = \frac{8}{15}$. (In general, $(\frac{a}{b}) \times (\frac{c}{d}) = \frac{(ac)}{(bd)}$. b. Find the area of a rectangle with fractional side lengths by tiling it with unit squares of the appropriate unit fraction side lengths, and show that the area is the same as would be found by multiplying the side lengths. Multiply fractional side lengths to find areas of rectangles, and represent fraction products as rectangular areas. c. Provide context/real-world situations when teaching multiplication of fractions. For example, if you want to half a recipe that requires $\frac{3}{4}$ cup of 1 sugar, how much sugar would you need?
Limitations	None
5.NF.5: Interpret multiplication as scaling (resizing).	
Clarifications	a. Compare the size of a product to the size of one factor on the basis of the size of the other factor, without performing the indicated multiplication. b. Explain why multiplying a given number by a fraction greater than 1 results in a product greater than the given number (recognizing multiplication by whole numbers greater than 1 as a familiar case); explain why multiplying a given number by a fraction less than 1 results in a product smaller than the given number; and relate the principle of fraction equivalence $\frac{a}{b} = \frac{(n \times a)}{(n \times b)}$ to the effect of multiplying $\frac{a}{b}$ by 1.
Limitations	None

5.NF.6: Solve real world problems involving multiplication of fractions and mixed numbers, e.g., by using visual fraction models or equations to represent the problem.	
Clarifications	None
Limitations	None
5.NF.7: Apply and extend previous understandings of division to divide unit fractions by whole numbers and whole numbers by unit fractions.	
Clarifications	a. Interpret division of a unit fraction by a non-zero whole number, and compute such quotients. For example, create a story context for $(\frac{1}{3}) \div 4$, and use a visual fraction model to show the quotient. Use the relationship between multiplication and division to explain that $(\frac{1}{3}) \div 4 = \frac{1}{12}$ because $(\frac{1}{12}) \times 4 = \frac{1}{3}$. b. Interpret division of a whole number by a unit fraction, and compute such quotients. For example, create a story context for $4 \div (\frac{1}{5})$, and use a visual fraction model to show the quotient. Use the relationship between multiplication and division to explain that $4 \div (\frac{1}{5}) = 20$ because $20 \times (\frac{1}{5}) = 4$. c. Solve real world problems involving division of unit fractions by non-zero whole numbers and division of whole numbers by unit fractions, e.g., by using visual fraction models and equations to represent the problem. For example, how much chocolate will each person get if 3 people share $\frac{1}{2}$ lb of chocolate equally? How many $\frac{1}{3}$ cup servings are in 2 cups of raisins?
Limitations	None

Measurement and Data

Convert like measurement units within a given measurement system.	
5.MD.1: Convert among different-sized standard measurement units within a given measurement system (e.g., convert 5 cm to 0.05 m), and use these conversions in solving multi-step, real world problems.	
Clarifications	None
Limitations	◦ Decimals limited to hundredths.

Represent and interpret data.	
5.MD.2: Make a line plot to display a data set of measurements in fractions of a unit ($\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$). Use operations on fractions for this grade to solve problems involving information presented in line plots. For example, given different measurements of liquid in identical beakers, find the amount of liquid each beaker would contain if the total amount in all the beakers were redistributed equally.	
Clarifications	None
Limitations	None
Geometric measurement: understand concepts of volume and relate volume to multiplication and to addition.	
5.MD.3: Recognize volume as an attribute of solid figures and understand concepts of volume measurement.	
Clarifications	a. A cube with side length 1 unit, called a "unit cube," is said to have "one cubic unit" of volume, and can be used to measure volume. b. A solid figure which can be packed without gaps or overlaps using n unit cubes is said to have a volume of n cubic units.
Limitations	None
5.MD.4: Measure volumes by counting unit cubes, using cubic cm, cubic in, cubic ft, and improvised units.	
Clarifications	None
Limitations	None
5.MD.5: Relate volume to the operations of multiplication and addition and solve real world and mathematical problems involving volume.	
Clarifications	a. Find the volume of a right rectangular prism with whole-number side lengths by packing it with unit cubes, and show that the volume is the same as would be found by multiplying the edge lengths, equivalently by multiplying the height by the area of the base. Represent threefold whole-number products as volumes, e.g., to represent the associative property of multiplication. b. Apply the formulas $V = l \times w \times h$ and $V = B \times h$ for rectangular prisms to find volumes of right rectangular prisms with whole-number edge lengths in the context of solving real world and mathematical problems. c. Recognize volume as additive. Find volumes of solid figures composed of two non-overlapping right rectangular prisms by adding the volumes of the non-overlapping parts, applying this technique to solve real world problems.
Limitations	None

Geometry

Graph points on the coordinate plane to solve real-world and mathematical problems.	
Cluster Limitations	◦ Limited to positive numbers in Quadrant 1.
5.G.1: Use a pair of perpendicular number lines, called axes, to define a coordinate system, with the intersection of the lines (the origin) arranged to coincide with the 0 on each line and a given point in the plane located by using an ordered pair of numbers, called its coordinates. Understand that the first number indicates how far to travel from the origin in the direction of one axis, and the second number indicates how far to travel in the direction of the second axis, with the convention that the names of the two axes and the coordinates correspond (e.g., x-axis and x-coordinate, y-axis and y-coordinate).	
Clarifications	None
Limitations	◦ See cluster-level limitation above.
5.G.2: Represent real world and mathematical problems by graphing points in the first quadrant of the coordinate plane, and interpret coordinate values of points in the context of the situation.	
Clarifications	None
Limitations	◦ See cluster-level limitation above.
Classify two-dimensional figures into categories based on their properties.	
Cluster Clarifications	◦ Properties can include angles, side lengths, symmetry, congruent parts, and whether there are parallel or perpendicular lines.
5.G.3: Understand that attributes belonging to a category of two-dimensional figures also belong to all subcategories of that category. For example, all rectangles have four right angles and squares are rectangles, so all squares have four right angles.	
Clarifications	a. See cluster-level clarification above.
Limitations	None
5.G.4: Classify two-dimensional figures in a hierarchy based on properties.	
Clarifications	a. See cluster-level clarification above.
Limitations	None

Grade 6

Ratios and Proportional Relationships

Understand ratio concepts and use ratio reasoning to solve problems.	
6.RP.1: Understand the concept of a ratio.	
Clarifications	a. Use ratio language to describe a ratio relationship between two quantities.
Limitations	◦ Limit to comparing three quantities.
6.RP.2: Understand the concept of a unit rate $\frac{a}{b}$ associated with a ratio a:b with $b \neq 0$, and use rate language in the context of a ratio relationship.	
Clarifications	None
Limitations	None
6.RP.3: Use multiple representations to show ratios and rate reasoning to solve real-world and mathematical problems.	
Clarifications	a. Make tables of equivalent ratios relating quantities with whole-number measurements, find missing values in the tables, and plot the pairs of values on the coordinate plane. Use tables to compare ratios. b. Solve unit rate problems with unit pricing and constant speed. c. Find a percent of a quantity as a rate per 100 (e.g., 30% of a quantity means $\frac{30}{100}$ times the quantity); solve problems involving finding the whole, given a part and the percent. d. Use ratio reasoning to convert measurement units within the same system and manipulate and transform units appropriately when multiplying or dividing quantities. If not converting within the same system, use the given conversion rate.
Limitations	◦ Not required to use complex fractions.

The Number System

Apply and extend previous understandings of multiplication and division to divide fractions by fractions.	
6.NS.1: Interpret and compute quotients of fractions, and solve word problems involving division of fractions by fractions, e.g., by using visual fraction models and equations to represent the problem.	
Clarifications	None
Limitations	None
Compute fluently with multi-digit numbers and find common factors and multiples.	
6.NS.2: Fluently divide multi-digit numbers within four decimal places using an algorithm.	
Clarifications	None
Limitations	◦ Does not require timed tests. ◦ Practice all operations with decimals whose solutions do not exceed 4 decimal places (e.g., 3.4325).
6.NS.3: Fluently add, subtract, multiply, and divide multi-digit decimals for each operation (solutions within four decimal places).	
Clarifications	None
Limitations	◦ Does not require timed tests. ◦ Practice all operations with decimals whose solutions do not exceed 4 decimal places (e.g., 3.4325).
6.NS.4: Find the greatest common factor of two whole numbers less than or equal to 100 and the least common multiple of two whole numbers less than or equal to 12. Use the distributive property to express a sum of two whole numbers 1–100 with a common factor as a multiple of a sum of two whole numbers with no common factor.	
Clarifications	None
Limitations	None

Apply and extend previous understandings of numbers to the system of rational numbers.	
6.NS.5: Understand that positive and negative numbers are used together to describe quantities having opposite directions or values; use positive and negative numbers to represent quantities in real-world contexts, explaining the meaning of 0 in each situation.	
Clarifications	None
Limitations	None
6.NS.6: Understand a rational number as a point on the number line. Extend number line diagrams and coordinate axes familiar from previous grades to represent points on the line and in the plane with negative number coordinates.	
Clarifications	a. Recognize opposite signs of numbers as indicating locations on opposite sides of 0 on the number line; recognize that the opposite of the opposite of a number is the number itself, e.g., –(–3) = 3, and that 0 is its opposite. b. Understand signs of numbers in ordered pairs as indicating locations in quadrants of the coordinate plane; recognize that when two ordered pairs differ only by signs, the locations of the points are related by reflections across one or both axes. c. Find and graph integers and benchmark rational numbers on a horizontal or vertical number line diagram. d. Find and graph pairs of integers and benchmark rational numbers on a coordinate plane.
Limitations	None
6.NS.7: Understand ordering and absolute value of rational numbers.	
Clarifications	a. Compare rational numbers and their relative position on a horizontal or vertical number line. For example, interpret -3 > -7 as a statement that -3 is located to the right of -7 on a number line oriented from left to right. b. Write, interpret, and explain statements of order for rational numbers in real-world contexts. c. Understand the absolute value of a rational number as its distance from 0 on the number line; interpret absolute value as magnitude for a positive or negative quantity in a real-world situation. d. Distinguish comparisons of absolute value from statements about order.
Limitations	None
6.NS.8: Solve real-world and mathematical problems by graphing points in all four quadrants of the coordinate plane. Include the use of coordinates and absolute value to find distances between points with the same first coordinate or the same second coordinate.	
Clarifications	None
Limitations	◦ The distance between points should either have the same first coordinate or the same second coordinate.

Expressions, Equations, and Inequalities

Apply and extend previous understandings of arithmetic to algebraic expressions.	
6.EEI.1: Write and evaluate numerical expressions involving whole-number exponents.	
Clarifications	None
Limitations	None
6.EEI.2: Write, read, and evaluate expressions in which letters represent numbers. Understand that these letters are called variables.	
Clarifications	a. Write expressions that record operations with numbers and with letters standing for numbers. b. Identify parts of an expression using mathematical terms (sum, term, product, factor, quotient, coefficient); view one or more parts of an expression as a single entity. c. Evaluate expressions at specific values of their variables. Include expressions that arise from formulas used in real-world problems. Perform arithmetic operations, including those involving whole-number exponents, in the conventional order when there are no parentheses to specify a particular order (Order of Operations).
Limitations	None
6.EEI.3: Apply the properties of operations to generate equivalent expressions.	
Clarifications	None
Limitations	None
6.EEI.4: Identify when two expressions are equivalent (i.e., when the two expressions name the same number regardless of which value is substituted into them).	
Clarifications	None
Limitations	None
Reason about and solve one-variable equations.	
6.EEI.5: Understand solving an equation as a process of answering a question: which values from a specified set, if any, make the equation true? Use substitution to determine whether a given number in a specified set makes an equation true.	
Clarifications	None
Limitations	None

6.EEI.6: Use variables to represent numbers and write expressions when solving a real-world or mathematical problem; understand that a variable can represent an unknown number, or, depending on the purpose at hand, any number in a specified set.	
Clarifications	None
Limitations	None
6.EEI.7: Solve real-world and mathematical problems by writing and solving equations of the form $x + p = q$ and $px = q$ for cases in which p, q, and x are all nonnegative rational numbers.	
Clarifications	None
Limitations	None
Represent and analyze quantitative relationships between dependent and independent variables.	
6.EEI.8: Use variables to represent two quantities in a real-world problem that change in relationship to one another.	
Clarifications	a. Write an equation to express one quantity, thought of as the dependent variable, in terms of the other quantity, thought of as the independent variable. b. Analyze the relationship between the dependent and independent variables using graphs and tables, and relate these to the equation.
Limitations	None

Geometry

Solve real-world and mathematical problems involving area, surface area, and volume.	
6.G.1: Find the area of right triangles, other triangles, special quadrilaterals, and polygons by composing into rectangles or decomposing into triangles and other shapes; apply these techniques within the context of real-world and mathematical problems.	
Clarifications	None
Limitations	◦ Memorize the basic formulas for only rectangles and triangles.

6.G.2: Find the volume of a right rectangular prism with fractional edge lengths by packing it with unit cubes of the appropriate unit fraction edge lengths, and show that the volume is the same as is found by multiplying the edge lengths of the prism. Apply the formulas $V = lwh$ and $V = Bh$ to volumes of right rectangular prisms with fractional edge lengths in the context of solving real-world and mathematical problems.	
Clarifications	None
Limitations	None
6.G.3: Draw polygons in the coordinate plane given coordinates for the vertices; use coordinates to find the length of a side joining points with the same first coordinate or the same second coordinate. Apply these techniques in the context of solving real-world and mathematical problems.	
Clarifications	None
Limitations	None
6.G.4: Represent three-dimensional figures using nets made up of rectangles and triangles, and use the nets to find the surface area of these figures. Apply these techniques in the context of solving real-world and mathematical problems.	
Clarifications	None
Limitations	◦ Memorization of surface area formulas is not required.

Data Literacy: Statistics and Probability

Understand how to interpret statistical data.	
6.DL.1: Recognize a statistical question as one that anticipates variability in the data related to the question and accounts for it in the answers.	
Clarifications	None
Limitations	None
6.DL.2: Understand that a set of data collected to answer a statistical question has a distribution which can be described by its center, spread, and overall shape.	
Clarifications	None
Limitations	None

6.DL.3: Recognize that a measure of center (mean, median, and mode) for a numerical data set summarizes all of its values with a single number, while a measure of variation (range and IQR) describes how its values vary with a single number.	
Clarifications	None
Limitations	None
Represent and interpret data using distributions.	
6.DL.4: Display numerical data in plots on a number line, including dot plots, histograms, and box plots.	
Clarifications	None
Limitations	None
6.DL.5: Summarize numerical data sets in relation to their context.	
Clarifications	a. Report the number of observations. b. Describe the nature of the attribute under investigation, including how it was measured and its units of measurement. c. Give quantitative measures of center (median and/or mean) and variability (interquartile range), as well as describe any overall pattern and any striking deviations from the overall pattern with reference to the context in which the data were gathered. d. Relate the choice of measures of center and variability to the shape of the data distribution and the context in which the data were gathered.
Limitations	None

Grade 7

Ratios and Proportional Relationships

Analyze proportional relationships and use them to solve real-world and mathematical problems.	
7.RP.1: Compute unit rates associated with ratios of fractions, including ratios of lengths, areas, and other quantities measured in like or different units.	
Clarifications	None
Limitations	None
7.RP.2: Recognize and represent proportional relationships between quantities.	
Clarifications	a. Decide whether two quantities are in a proportional relationship. b. Identify the constant of proportionality (unit rate) in tables, graphs, equations, diagrams, and verbal descriptions of proportional relationships. c. Represent proportional relationships by equations. d. Explain what a point (x, y) on the graph of a proportional relationship means in terms of the situation, with special attention to the points $(0, 0)$ and $(1, r)$ where r is the unit rate.
Limitations	None
7.RP.3: Use proportional relationships to solve multistep ratio and percent real-world problems.	
Clarifications	None
Limitations	◦ Limit to simple interest, tax, markups and markdowns, gratuities and commissions, percent increase and decrease, and percent error.

The Number System

Apply and extend previous understandings of operations with fractions to add, subtract, multiply, and divide rational numbers.	
7.NS.1: Apply and extend previous understandings of addition and subtraction to add and subtract rational numbers; represent addition and subtraction on a horizontal or vertical number line diagram.	
Clarifications	a. Describe situations in which opposite quantities combine to make 0. b. Understand $p + q$ as the number located a distance $\lvert q \rvert$ from p, in the positive or negative direction depending on whether q is positive or negative. Show that a number and its opposite have a sum of 0 (are additive inverses). Interpret sums of rational numbers by describing real-world contexts. c. Understand subtraction of rational numbers as adding the additive inverse, $p - q = p + (-q)$. Show that the distance between two rational numbers on the number line is the absolute value of their difference, and apply this principle in real-world contexts. d. Apply properties of operations as strategies to add and subtract rational numbers.
Limitations	None
7.NS.2: Apply and extend previous understandings of multiplication and division and of fractions to multiply and divide rational numbers.	
Clarifications	a. Understand that multiplication is extended from fractions to rational numbers by requiring that operations continue to satisfy the properties of operations, particularly the distributive property and the rules for multiplying signed numbers. Interpret products of rational numbers by describing real-world contexts. b. Understand that integers can be divided, provided that the divisor is not zero, and every quotient of integers (with non-zero divisors) is a rational number. If p and q are integers, then $-(\frac{p}{q}) = \frac{-p}{q} = \frac{p}{-q}$. Interpret quotients of rational numbers by describing real-world contexts. c. Apply properties of operations as strategies to multiply and divide rational numbers. d. Convert a rational number to a decimal using long division to four decimal places; know that the decimal form of a rational number terminates in 0s or eventually repeats.
Limitations	◦ Limit to four decimal places for calculating by long division. ◦ Limit to rational numbers that repeat within four decimal places so it can be determined by long division whether a number terminates or eventually repeats.
7.NS.3: Solve real-world and mathematical problems involving the four operations with rational numbers.	
Clarifications	a. Perform computations with complex fractions.
Limitations	◦ Rational numbers and complex fractions can be used, but should contain fractions with single-digit numerators and denominators. ◦ Limit to rational numbers that repeat within four decimal places so it can be determined by long division whether a number terminates or eventually repeats.

Expressions, Equations, and Inequalities

Use properties of operations to generate equivalent expressions.	
7.EEI.1: Apply properties of operations as strategies to add, subtract, and expand linear expressions with rational coefficients.	
Clarifications	None
Limitations	None
7.EEI.2: Understand that rewriting an expression in different forms in a problem context can shed light on the problem and how the quantities in it are related.	
Clarifications	None
Limitations	None
Solve real-life and mathematical problems using numerical and algebraic expressions and equations.	
7.EEI.3: Solve multi-step real-world and mathematical problems posed with positive and negative rational numbers in any form (whole numbers, fractions, and decimals), using tools strategically.	
Clarifications	a. Apply properties of operations to calculate numbers in any form. b. Convert between different forms of rational numbers. c. Assess the reasonableness of answers using mental computation and estimation strategies.
Limitations	None
7.EEI.4: Use variables to represent quantities in a real-world or mathematical problem, and construct simple equations to solve problems by reasoning about the quantities.	
Clarifications	a. Solve word problems leading to equations of the form $px + q = r$ and $p(x + q) = r$, where p, q, and r are specific rational numbers. Solve equations of these forms fluently. b. Compare an algebraic solution to an arithmetic solution, identifying the sequence of the operations used in each approach.
Limitations	None

Geometry

Understand and solve problems involving scale drawings.	
7.G.1: Solve problems involving scale drawings of geometric figures, including computing actual lengths and areas from a scale drawing and reproducing a scale drawing at a different scale.	
Clarifications	None
Limitations	None
Solve real-world and mathematical problems involving angle measure, area, surface area, and volume.	
7.G.2: Know the formulas for the area and circumference of a circle and use them to solve problems.	
Clarifications	a. Give an informal derivation of the relationship between circumference and area of a circle. b. Solve real-world problems using circles and circumference.
Limitations	◦ Limit equations to the forms $px + q = r$ or $p(x + q) = r$, where p, q, and r are rational numbers.
7.G.3: Given the formula, solve real-world and mathematical problems involving area, volume, and surface area of two- and three-dimensional objects composed of circles, triangles, quadrilaterals, polygons, cubes, and right prisms.	
Clarifications	None
Limitations	None

Data Literacy: Statistics and Probability

Use random sampling to draw inferences about a population.	
7.DL.1: Understand that statistics can be used to gain information about a population by examining a sample of the population; generalizations about a population from a sample are valid only if the sample is representative of that population. Understand that random sampling tends to produce representative samples and support valid inferences.	
Clarifications	None
Limitations	None

7.DL.2: Use data from a random sample to draw inferences about a population with an unknown characteristic of interest. Generate multiple samples (or simulated samples) of the same size to gauge the variation in estimates or predictions.	
Clarifications	None
Limitations	None
Represent and interpret comparative inferences about two populations.	
Cluster Limitations	◦ Does not require calculating mean absolute deviation.
7.DL.3: Informally assess the degree of visual overlap of two numerical data distributions with similar variabilities, measuring the difference between the centers by expressing it as a multiple of a measure of variability (range and IQR).	
Clarifications	None
Limitations	◦ See cluster-level limitation above.
7.DL.4: Given a data set from a random sample, use measures of center (mean, median, and mode) and measures of variability (range and IQR) for numerical data to draw informal comparative inferences about two populations.	
Clarifications	None
Limitations	◦ See cluster-level limitation above.
Investigate chance processes and develop, use, and evaluate probability models.	
7.DL.5: Understand that the probability of a chance event is a number between 0 and 1 that expresses the likelihood of the event occurring. Larger numbers indicate a greater likelihood. A probability near 0 indicates an unlikely event, a probability around $\frac{1}{2}$ indicates an event that is neither unlikely nor likely, and a probability near 1 indicates a likely event.	
Clarifications	None
Limitations	None
7.DL.6: Approximate the probability of a chance event by collecting data on the chance process that produces it and observing its long-run relative frequency, and predict the approximate relative frequency given the probability.	
Clarifications	None
Limitations	None

7.DL.7: Develop a probability model and use it to find probabilities of events. Compare probabilities from a model to observed frequencies; if the agreement is not good, explain possible sources of the discrepancy.	
Clarifications	a. Develop a uniform probability model by assigning equal probability to all outcomes, and use the model to determine probabilities of events. b. Develop a probability model (which may not be uniform) by observing frequencies in data generated from a chance process.
Limitations	None
7.DL.8: Find probabilities of compound events using organized lists, tables, tree diagrams, and simulation.	
Clarifications	a. Understand that, just as with simple events, the probability of a compound event is the fraction of outcomes in the sample space for which the compound event occurs. b. Represent sample spaces for compound events using methods such as organized lists, tables, and tree diagrams. For an event described in everyday language (e.g., “rolling double sixes”), identify the outcomes in the sample space that compose the event. c. Design and use a simulation to generate frequencies for compound events.
Limitations	None

Grade 8

The Number System

Know that there are numbers that are not rational, and approximate them by rational numbers.	
8.NS.1: Determine whether a number is rational or irrational.	
Clarifications	None
Limitations	None
8.NS.2: Use rational approximations to compare the size of irrational numbers, estimate their value, and locate them approximately on a number line.	
Clarifications	None
Limitations	None

Linear Relationships

Understand the connections between proportional relationships, lines, and linear equations.	
Cluster Limitations	◦ Not required to know (but should be exposed to the idea that) linear relationships come under a larger umbrella of a special kind of relationship called functions. ◦ Not expected to determine whether or not a relationship is a function. ◦ Not required to know function notation.
8.LR.1: Graph proportional relationships from multiple representations (tables, equations, and/or verbal descriptions). Interpret the unit rate as the slope of the graph in a proportional relationship.	
Clarifications	a. Compare two different proportional relationships represented differently.
Limitations	◦ See cluster-level limitations above.

8.LR.2: Write an equation representing a linear relationship between two quantities.	
Clarifications	a. Determine the rate of change and initial value from a table, graph, equation, or verbal description. b. Interpret the rate of change and initial value from a table, graph, equation, or verbal description.
Limitations	◦ See cluster-level limitations above.
8.LR.3: Explain that the slope m is the same between any two distinct points on a non-vertical line in a coordinate plane because they grow by equal differences over equal intervals.	
Clarifications	None
Limitations	◦ See cluster-level limitations above.
8.LR.4: Compare the properties between two linear relationships, each represented in different ways (tables, graphs, equations, and verbal description).	
Clarifications	None
Limitations	◦ See cluster-level limitations above.
8.LR.5: Interpret the equation $y = mx + b$ as a linear relationship, whose graph is a straight line. Give examples of equations that are not linear.	
Clarifications	None
Limitations	◦ See cluster-level limitations above.
8.LR.6: Explain what a point (x, y) on the graph of a linear relationship means in terms of the situation.	
Clarifications	None
Limitations	◦ See cluster-level limitations above.
Non-linear relationships between quantities.	
8.LR.7: Describe qualitatively the relationship between two quantities by analyzing a graph (e.g., where the graph is increasing or decreasing, linear or nonlinear). Sketch a graph that exhibits the qualitative features that have been described verbally.	
Clarifications	None
Limitations	None

Expressions, Equations, and Inequalities

Work with radicals and integer exponents.	
8.EEI.1: Know and apply the properties of integer exponents to generate equivalent numerical expressions.	
Clarifications	None
Limitations	None
8.EEI.2: Use square root and cube root symbols to represent solutions to equations of the form $x^2 = p$ and $x^3 = p$, where p is a positive rational number. Evaluate square roots of small perfect squares and cube roots of small perfect cubes. Know that $\sqrt{2}$ is irrational.	
Clarifications	None
Limitations	None
8.EEI.3: Use numbers expressed in the form of a single digit times an integer power of 10 to estimate very large or very small quantities, and to express how many times as much one is than the other.	
Clarifications	None
Limitations	None
8.EEI.4: Use scientific notation and choose units of appropriate size to represent very large or very small quantities. Interpret scientific notations that have been generated by technology.	
Clarifications	None
Limitations	◦ Not expected to compute with scientific notation.
Analyze and solve equations and inequalities with one variable.	
Cluster Limitations	◦ Use reasonable rational numbers that will produce reasonable solutions.
8.EEI.5: Solve linear equations in one variable.	
Clarifications	a. Give examples of linear equations in one variable with one solution, an infinite number of solutions, and no solutions. b. Solve linear equations with rational number coefficients, including equations whose solutions require expanding expressions using the distributive property and combining like terms.
Limitations	◦ See cluster-level limitation above.

8.EEI.6: Write an inequality of the form $x > c$ or $x < c$ to represent a constraint or condition in a real-world or mathematical problem. Recognize that inequalities of the form $x > c$ or $x < c$ have infinitely many solutions; represent solutions of such inequalities on number line diagrams.	
Clarifications	a. Use variables to represent quantities in a real-world or mathematical problem, and construct simple inequalities to solve problems by reasoning about the quantities. b. Solve word problems leading to inequalities of the form $px + q > r$ or $px + q < r$, where p, q, and r are specific rational numbers. Graph the solution set of the inequality and interpret it in the context of the problem.
Limitations	◦ See cluster-level limitation above.
Analyze and solve equations with two variables.	
8.EEI.7: Analyze and explain the solutions of simultaneous linear equations using graphs, tables, and by inspection.	
Clarifications	a. Describe and explain that solutions to a system of two linear equations in two variables correspond to points of intersection on a graph or table, because points of intersection satisfy both equations simultaneously. b. Solve systems of two linear equations in two variables using tables, by graphing or by inspection for simple cases. c. Solve real-world and mathematical problems leading to two linear equations in two variables using tables, by graphing, or by inspection for simple cases.
Limitations	◦ Not required to solve systems of equations algebraically.

Geometry

Understand congruence and similarity using manipulatives or geometry software.	
8.G.1: Describe the properties of rigid transformations (rotations, reflections, and translations) and non-rigid transformations (dilations).	
Clarifications	a. Analyze two-dimensional shapes to identify and construct lines of symmetry.
Limitations	None
8.G.2: Describe the difference between congruence and similarity.	
Clarifications	None
Limitations	None

Solve real-world and mathematical problems involving angle measurements.	
8.G.3: Identify supplementary, complementary, vertical, and adjacent angles.	
Clarifications	None
Limitations	None
8.G.4: Use facts about supplementary, complementary, vertical, and adjacent angles in a multi-step problem to write and solve equations for an unknown angle in a figure.	
Clarifications	None
Limitations	◦ Not required to know transversals. ◦ Not required to know angle bisectors.
8.G.5: Apply supplementary, complementary, vertical, and adjacent angles in real-world contexts.	
Clarifications	None
Limitations	None
Solve real-world and mathematical problems involving sectors and arc length.	
Cluster Limitations	◦ Not expected to memorize the formulas for sectors and arc lengths. ◦ Work with benchmark fractions: $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$. ◦ Work with benchmark degrees: 180°, 90°, 45°, 22.5°.
8.G.6: Understand the relationship between the area of the circle and a sector.	
Clarifications	a. Solve for the area of a sector of a circle when provided with the radius and the central angle in degrees.
Limitations	◦ See cluster-level limitations above.
8.G.7: Understand the relationship between the circumference of the circle and arc length.	
Clarifications	a. Solve for the arc length of a circle when provided with the radius and the central angle in degrees. b. Solve real-world and mathematical problems involving sectors and arc lengths.
Limitations	◦ See cluster-level limitations above.

Solve real-world and mathematical problems involving the volume of cylinders, cones, and spheres.	
8.G.8: Solve real-world and mathematical problems involving volume, given the formulas for the volume of cylinders, cones, and spheres.	
Clarifications	None
Limitations	◦ Not expected to memorize the formulas for cylinders, cones, and spheres (formuals will be provided). ◦ Not required to calculate the slant height of the cone.

Data Literacy: Statistics and Probability

Represent and interpret patterns of association in bivariate data.	
8.DL.1: Construct scatter plots for bivariate measurement data to investigate patterns of association between two quantities.	
Clarifications	a. Describe and interpret patterns such as clustering, outliers, positive or negative association, linear association, and nonlinear association.
Limitations	None
8.DL.2: Determine if a relationship between two quantities can be modeled with a straight line, then informally fit a straight line.	
Clarifications	None
Limitations	◦ Not required to know formal line of best fit/regression.
8.DL.3: Estimate a line that could model the relationship between two quantitative variables to solve problems, interpreting the slope and intercept.	
Clarifications	None
Limitations	None

8.DL.4: Understand that patterns of association can also be seen in bivariate categorical data by displaying frequencies and relative frequencies in a two-way table.	
Clarifications	a. Interpret and create a two-way table given data and/or two variable categories. Describe possible associations between the two variables using relative frequencies calculated for rows and columns.
Limitations	◦ Not required to create a two-way frequency table from scratch.

Geometry + Probability

ʻŌpaʻa 1 - Exploring and Applying the Pythagorean Theorem

Students develop a comprehensive understanding of the Pythagorean Theorem, including working with proofs, and moving seamlessly between geometric, algebraic, and coordinate representations while flexibly representing equivalent expressions. They apply the theorem to find missing side lengths in right triangles, demonstrate algebraic manipulation, and connect geometric representations to equations. Students engage in sense-making throughout, interpreting solutions within contextual scenarios, and justifying reasoning through various representations.	
G.1.1: Explain proofs of the Pythagorean Theorem and its converse.	
Clarifications	a. Proofs should include a variety of methods (e.g., area-based, algebraic, and dissection models).
Limitations	◦ Proficiency is demonstrated by reasoning through and explaining the Pythagorean Theorem and its converse. Formal two-column proofs are optional.
G.1.2: Apply the Pythagorean Theorem to determine unknown side lengths of right triangles and other polygons in real-world and mathematical problems.	
Clarifications	a. Include special right triangles (i.e., “45° – 45° – 90°” triangles and “30° – 60° – 90°” triangles). b. Use the Pythagorean Theorem to find the perimeter of polygons and the area of triangles, rectangles, or polygons that can be deconstructed into other shapes (triangles, rectangles, etc).
Limitations	◦ Use exact values for mathematical problems (do not simplify radicals) and approximate values for contextual situations.
G.1.3: Apply the Pythagorean Theorem to find the distance between two points on a coordinate graph.	
Clarifications	a. Use the Pythagorean Theorem and the distance formula in the coordinate plane to compute measures of polygons.
Limitations	None

ʻŌpaʻa 2 - Constructing and Applying Arguments About Lines and Triangles

Students construct and communicate geometric arguments through proofs, logical reasoning, and geometric/spatial/visual understanding. They demonstrate representational flexibility by translating between visual diagrams, symbolic notation, and verbal descriptions of geometric relationships. They construct arguments using visualizable relationships between geometric figures, such as lines and angle measures. They demonstrate strong reasoning and sensemaking as they apply theorems and justify arguments related to triangles to solve problems. They use technological tools to enhance understanding and problem-solving and to support their work when appropriate.	
G.2.1: Construct and prove relationships between parallel and perpendicular lines using coordinate geometry.	
Clarifications	a. Construct parallel and perpendicular lines through a given point, as well as perpendicular and angle bisectors. b. Use coordinate geometry to show that two lines are parallel or perpendicular by comparing their slopes, and apply this information to solve problems.
Limitations	◦ Equations should be written in slope-intercept form.
G.2.2: Justify line and angle relationships and use them to solve problems.	
Clarifications	a. Angle relationships should include vertical angles and relationships formed when a transversal crosses parallel lines (e.g., alternate interior and corresponding angles, and their converses). b. Relationships involving lines and angles include: supplementary & complementary angles, perpendicular bisector and angle bisector. c. Points on a perpendicular bisector of a line segment are exactly those equidistant from the segment's endpoints.
Limitations	None
G.2.3: Justify theorems of triangles and use them to solve problems.	
Clarifications	a. Theorems include: Triangle Sum Theorem and Isosceles Triangle Theorem.
Limitations	None

ʻŌpaʻa 3 - Exploring and Applying Rigid Motion Transformations and Congruence

Students continue to use geometric figures as a foundation for sharpening their own sensemaking, reasoning, and construction of viable arguments. They define congruence in terms of rigid transformations (translations, rotations, and reflections) and apply that definition to create mathematical arguments, solve problems, and explore situations. They explore and analyze various forms of symmetry, including reflective and rotational symmetry, and how these relate to transformations. Students use multiple representations to describe congruence, using visual, algebraic, and coordinate-based methods. They use technology to foster their own sensemaking as they dynamically examine and reason logically about geometric transformations.	
G.3.1: Develop and apply definitions of translations, rotations and reflections of geometric figures.	
Clarifications	a. Perform a reflection about a specific line. b. Perform a rotation about a specified point. c. Perform a translation.
Limitations	None
G.3.2: Represent transformations as functions that take points in the plane as inputs and give other points as outputs.	
Clarifications	a. Interpret transformations in symbolic form.
Limitations	None
G.3.3: Use rigid motions to show that two figures are congruent.	
Clarifications	a. Compare transformations that use rigid and non-rigid motion. b. Use definitions to identify lines and points of symmetry and angles of rotation to map a figure onto itself. c. Explain triangle congruence (ASA, SAS, and SSS) using rigid motion transformations.
Limitations	◦ Proficiency is demonstrated by reasoning through and explaining how rigid motions justify triangle congruence. Formal two-column proofs are optional.

ʻŌpaʻa 4 - Exploring and Applying Similarity Transformations and Triangle Similarity

Students define similarity in terms of combinations of rigid transformations paired with dilations. They move between different ways of visualizing and expressing similarity (e.g., diagrams, coordinates, symbolic notation), and articulate the key differences and similarities between rigid and similarity transformations. Students determine whether two given figures are similar using properties of dilations and the center and scale factor. They use technology to explore and reason logically about geometric transformations.	
G.4.1: Understand and use the definition of similarity in terms of similarity transformations.	
Clarifications	a. Understand that a two-dimensional figure is similar to another if the second can be obtained from the first by a sequence of rotations, reflections, translations, and dilations.
Limitations	None
G.4.2: Understand and use triangle similarity theorems in contextual and mathematical situations.	
Clarifications	a. Include Angle-Angle, Side-Angle-Side, and Side-Side-Side. b. Determine whether or not two triangles are similar and use similarity to solve for missing values in similar triangles.
Limitations	◦ Proficiency is demonstrated by reasoning through and explaining how similarity transformations justify triangle similarity. Formal two-column proofs are optional.
G.4.3: Use congruence and similarity criteria for triangles to solve problems and to prove relationships in geometric figures.	
Clarifications	a. Determine unknown lengths of segments of geometric figures and apply concepts of congruence and similarity to determine measures and solve problems. b. Know that a line parallel to one side of a triangle that intersects the other two sides divides those two sides proportionally and use that knowledge to find unknown values.
Limitations	◦ Proficiency is demonstrated by reasoning through and explaining how rigid motions and/or similarity transformations justify triangle congruence and similarity. Formal two-column proofs are optional.

ʻŌpaʻa 5 - Exploring and Applying Right Triangle Trigonometric Ratios

Building on their previous knowledge of triangles, students explore right triangle trigonometry, focusing on trigonometric ratios as properties of angles in a triangle. They use these ratios, along with the Pythagorean Theorem, to solve problems in both real-world and mathematical contexts. Further, they derive the equation of a circle given its center and radius, demonstrating a connection between the Pythagorean Theorem and the algebraic representation of circles.	
G.5.1: Use trigonometric ratios to find unknown values in right triangles.	
Clarifications	a. Use the lengths of the sides of a triangle and the relationship between the sine and cosine of complementary angles to find one given the other. b. Explain why trigonometric ratios remain constant in right triangles with the same angles, regardless of the size of those triangles.
Limitations	None
G.5.2: Solve contextual and mathematical problems that require the use of a trigonometric ratio and the Pythagorean Theorem.	
Clarifications	a. Include applications to write an equation of a circle given the center and radius. b. Include angle of elevation and angle of depression.
Limitations	◦ Limit trigonometry to right triangles; do not include Law of Sines and Law of Cosines. ◦ Limit to the sine, cosine, tangent ratios. ◦ Refrain from completing the square. ◦ Use $(x-h)^2+(y-k)^2=r^2$ for the equation of a circle.

ʻŌpaʻa 6 - Solving Problems and Interpreting Solutions Involving Area and Volume

Students solve problems and interpret solutions related to the area and volume of prisms, cylinders, pyramids, cones, and spheres, and demonstrate representational flexibility (e.g., diagrams, equations, verbal descriptions). They apply these skills to real-world situations by quantifying and modeling physical objects and phenomena. Students visualize and analyze relationships between two-dimensional and three-dimensional objects and demonstrate strong geometric, spatial, and visual reasoning skills.	
G.6.1: Use volume and surface area formulas for prisms, cylinders, pyramids, cones, and spheres to solve problems and apply to authentic contexts.	
Clarifications	a. Explain why the volume of a prism is $V = Bh$ (where B = area of the base). b. Apply the formulas for surface area and volume to solve problems in real-world contexts. c. Use the Pythagorean Theorem, special right triangles, and trigonometric ratios to find the height or slant height of a cone or pyramid in order to determine the volume and surface area.
Limitations	None
G.6.2: Visualize and identify relationships between two-dimensional and three-dimensional objects.	
Clarifications	a. Identify the parts of a polyhedron (faces, vertices, edges). b. Identify two-dimensional cross-sections of three-dimensional objects. c. Identify shapes of objects generated by rotations of two-dimensional objects.
Limitations	None
G.6.3: Apply the concepts of density based on area and volume in modeling situations.	
Clarifications	a. Use the concepts of density (mass per unit volume or other relevant measure per unit area or volume) to model real-world scenarios.
Limitations	None

ʻŌpaʻa 7 - Understanding and Applying Probability

Students extend their learning of area and volume to provide the groundwork for probability. They use these same geometric measures to determine the likelihood of events, shifting from counting discrete outcomes to understanding continuous possibilities through geometric probability. Students develop a conceptual understanding of probability through exploratory simulations of contextual scenarios (including technology). Students focus on understanding solutions to draw meaningful conclusions from data sets to bridge between experimental and theoretical probability. To create sample spaces, they use modeling strategies and apply probabilistic reasoning to interpret the results. They explore independence and conditional probability and develop representational flexibility by translating between different representations (e.g., tables, graphs, equations, simulations).	
G.7.1: Use simulations to estimate probabilities.	
Clarifications	a. Determine the likelihood of an event occurring (on scale of 0 to 1). b. Understand that the experimental probability gets closer to the actual probability as the number of trials increases.
Limitations	None
G.7.2: Define an appropriate sample space for an experiment and describe the possible outcomes for a situation as subsets of a sample space.	
Clarifications	a. Find the outcomes of situations that include words such as *and*, *or*, *not*, *if*, and *all*.
Limitations	None
G.7.3: Understand and explain the concepts of conditional probability and independence.	
Clarifications	a. Understand that two events A and B are independent if the probability of A and B occurring together is the product of their probabilities; i.e., $P(A \text{ and } B) = P(A) \cdot P(B)$. b. Understand the conditional probability of A given B as $\frac{P(A \text{ and } B)}{P(B)}$. c. Interpret independence of A and B as saying that the conditional probability of A given B is the same as the probability of A; i.e., $P(A\|B) = P(A)$. Similarly, the conditional probability of B given A is the same as the probability of B, i.e., $P(B\|A) = P(B)$.
Limitations	None

ʻŌpaʻa 8 - Understanding Counting and Expected Value

Students expand on their understanding of sample spaces by exploring counting principles including differentiating situations involving permutations versus combinations. Through this work, students are introduced to computing factorials and understanding their properties. Students develop an understanding of and use expected value to solve problems.	
G.8.1: Compute factorials and understand their properties.	
Clarifications	a. Use factorials to count arrangements with no replacement. b. Use technology for computing factorials when there are more than 5 items to arrange.
Limitations	None
G.8.2: Determine when a counting situation requires permutations or combinations and choose the appropriate method to solve real-world problems.	
Clarifications	a. Use permutations and combinations to compute the size of event space or sample space, and use this to compute probabilities. b. Use technology for computing permutations and combinations when the result is greater than 250.
Limitations	None
G.8.3: Calculate an expected value and use it to support a decision.	
Clarifications	a. Calculate expected values in real-world contexts. b. Use expected value to decide between two options that involve probabilities.
Limitations	◦ Discrete random variables should not have more than 5 outcomes.

Algebra 1 + Data Analytics

ʻŌpaʻa 1 - Working with a Single Variable

Students represent changing quantities with a single variable and write expressions to model real-world contexts. They generate equivalent expressions to reveal specific properties and use these expressions to write equations and inequalities that allow them to determine unknown values and solve problems.	
A1.1.1: Interpret parts of a linear expression, such as terms, constants, and coefficients that represent a quantity in terms of its context.	
Clarifications	a. Expressions should include the use of rational numbers as constants and coefficients; with only one variable of degree one: ◦ $3x + 2x$ ◦ $\frac{2}{3}m - 10$ ◦ $-3 - 0.25a$
Limitations	◦ Expressions should not have more than one variable or degrees other than one, such as: $3x + 2y$ or x^2–9.
A1.1.2: Construct a viable argument to justify the steps to reach a solution.	
Clarifications	a. Include various types of arguments (e.g., arrow diagrams, flow diagrams, algebra tiles, standard abstract representations).
Limitations	◦ Does not need to state formally the properties used for each step in solving an equation or inequality (e.g., state that “5” was added to both sides to balance the equation instead of stating the Addition Property of Equality principle).
A1.1.3: Create equations and inequalities in one variable and use them to solve problems.	
Clarifications	a. Determine the rate of change directly from the description of the situation.
Limitations	◦ Does not need to compute the rate of change from the description of the situation (e.g., rideshare charged $5 for 1 mile and $9 for 3 miles. How much would it cost to go 5 miles?).

ʻŌpaʻa 2 - Introducing Functions

Students explore the concept of functions and rates of change by analyzing and interpreting time vs. distance graphs. They describe and represent key points and intervals both verbally and symbolically, explaining their meaning in context. Students then apply their understanding of functions to represent and analyze bivariate data to make predictions.	
A1.2.1: Describe a given story/scenario graph showing a relationship between two quantities.	
Clarifications	a. Focus on bivariate data and analyzing relationships between two quantitative variables using line graphs.
Limitations	◦ Equations for piecewise functions should not be introduced/given. Example: $$f(x) = \begin{cases} 2x + 1, & 1 < x \leq 3 \\ x^2, & x > 3 \end{cases}$$
A1.2.2: Graph story/scenario functions by hand in simple cases between two quantities (limited to increasing, decreasing, constant, etc.) at different domain intervals in the graph.	
Clarifications	a. Focus on bivariate data and analyzing relationships between two quantitative variables using line graphs.
Limitations	None
A1.2.3: Represent data on two quantitative variables on a graph, and describe how the variables are related.	
Clarifications	a. Use verbal descriptions to describe the linear relationships between the input and output values in a graph.
Limitations	None
A1.2.4: Understand a function as a rule that assigns a unique output for every input and that functions model situations where one quantity determines another.	
Clarifications	a. Justify whether a relation is a function or not a function from a graph, table, and context (e.g., “This is a function because for any input (time), a person can have only one output (location).”)
Limitations	None

A1.2.5: Use function notation and interpret statements that use function notation in terms of the context and the relationship it describes.	
Clarifications	a. Evaluate a function given its equation and identify the inputs and outputs. For functions $f(x) = y$ that represent a context, interpret functional values in terms of the context. b. Determine whether a relation is a function or not a function from a graph, table, and context. c. The notation "$f(x)$" is read as "f of x"; "f" is the name of a function, "x" represents an input value in the domain of the function, and "$f(x)$" represents an output value in the range of the function corresponding to the input value x. d. Once a relationship is acknowledged as a function, y and $f(x)$ are interchangeable, requiring familiarity with both forms of notation.
Limitations	None

ʻŌpaʻa 3 - Representing and Using Linear Functions & Equations

Students extend their understanding of functions by examining linear functions characterized by a constant rate of change. They deepen their understanding of algebraic equivalence by using numeric, graphical, and symbolic representations to verify or refute equivalence statements as it pertains to linear functions. Students recognize that an equation poses a question, and its solution is the value that makes the statement true. Building on this, they explore linear inequalities and represent them in the coordinate plane. Students analyze and create linear systems to model real-world situations, find solutions, and interpret those solutions in context.	
A1.3.1: Use technology to create a linear regression model for a data set, and determine whether a linear model may or may not be appropriate.	
Clarifications	a. For data that shows a linear relationship, interpret the parameters (e.g., slope, intercepts) in terms of the context. b. Data sets should include linear and non-linear relationships to develop a working understanding of linear relationships in comparison to non-linear examples.
Limitations	◦ Avoid the use of data sets that may result in bivariate outliers (e.g. Age vs. Height, Temperature vs. Shaved Ice Sales). Instead, use variables that support linear trends and relationships (e.g. Height vs. Stride Length, Height vs. Shoe Size).

<table>
<tr><td colspan="2">A1.3.2: Create an equation that represents a linear relationship between two quantities.</td></tr>
<tr><td>Clarifications</td><td>a. Write an equation for a linear function given:
a verbal description between two quantities.
the coordinates of a point and the rate of change.
the coordinates of two points.
the graph of the function.</td></tr>
<tr><td>Limitations</td><td>Does not include equations for vertical lines, since this section focuses on linear functions.</td></tr>
<tr><td colspan="2">A1.3.3: Graph linear functions expressed symbolically and show key features of the graph, by hand or using technology. Interpret key features in terms of quantities in a context.</td></tr>
<tr><td>Clarifications</td><td>a. Identify the domain and range of a function given as a graph and/or context.
b. When describing the domain and range, create verbal descriptions that include types of numbers (whole, integers, rational) in which the scenario makes sense.
c. Identify the intercepts, end behavior, initial value, rate of change from the graph and interpret the meaning of those key features within the given context.</td></tr>
<tr><td>Limitations</td><td>When describing possible values within context, such as the number of weeks:
The possible quantities within the context should be described using phrases such as:
the number of weeks are whole numbers greater than or equal to zero
w is the number of weeks, $w \geq 0$
Descriptions of possible quantities within the context do not require more formalized notation such as:
set notation (e.g., $\{w : w \geq 0, w \in \text{whole numbers}\}$)
interval notation (e.g., $[0, +\infty)$)</td></tr>
<tr><td colspan="2">A1.3.4: Compare properties of two linear functions each represented in a different way (algebraically, graphically, numerically in tables, or by verbal descriptions).</td></tr>
<tr><td>Clarifications</td><td>a. Highlight the use of key features particularly initial value, rate of change when analyzing the linear functions.
b. Emphasize use of function notation.</td></tr>
<tr><td>Limitations</td><td>Does not include equations for vertical lines, since this section focuses on linear functions.
Focus on slope-intercept form.</td></tr>
</table>

A1.3.5: Graph the solutions to a linear equation/inequality in two variables.	
Clarifications	a. Graph the solutions to an equation in two variables as a line. ◦ Use algebraic properties of equality, make use of the structure of expressions within an equation, and use technology to solve equations. ◦ Be flexible between various forms of linear equations (e.g., slope-intercept, point-slope, standard). ◦ Graph linear equations for horizontal and vertical lines and recognize that horizontal lines are linear functions with rate of change 0 and vertical lines are not functions.
Limitations	None
A1.3.6: Create an equation for a line that is parallel or perpendicular to a given line passing through a given point.	
Clarifications	None
Limitations	◦ Coordinates of points should contain rational values. ◦ Slopes should be limited to rational values.
A1.3.7: Determine the intersection of two linear equations using substitution and graphing with technology. Interpret the meaning of the intersection in context.	
Clarifications	a. Use graphing technology to create a table of values and graph for the linear functions to find the point(s) of intersection, if any. ◦ Solve an equation of the form $f(x) = g(x)$ by graphing $y = f(x)$ and $y = g(x)$ and find where the graphs intersect.
Limitations	◦ When solving a system by hand, at least one of the equations should have a term with one of the variables with coefficient 1. ◦ Include: $2x + y = 4; x - 3y = 6$ ◦ Exclude: $2x + 5y = 10; 3x - 6y = 12$ ◦ When solving systems that do not have one equation with one of the variables with coefficient 1, use graphing technology. ◦ Elimination to solve a system should not be needed (elimination method will be covered in Algebra 2).
A1.3.8: Graph the solutions to a linear inequality system in two variables as the intersection of half-planes using technology.	
Clarifications	a. For contextual problems, consider the restrictions on the variable (e.g., number of items purchased must be positive) and write inequalities to model the situation. b. Systems with no solutions should also be considered.
Limitations	◦ Systems should include no more than 3 inequalities at a time.

ʻŌpaʻa 4 - Representing and Using Exponential Functions & Equations

Students extend their understanding of functions by examining exponential functions characterized by a constant ratio between successive values. They analyze key features of graphs to interpret the behavior of exponential growth and decay. Students also apply the properties of integer exponents to generate equivalent expressions. Students construct exponential functions from data, verbal descriptions, or graphs, and use these functions to estimate real-world solutions for a variety of inputs and outputs.	
A1.4.1: Know and apply the properties of integer exponents to generate equivalent numerical expressions.	
Clarifications	a. Explain why the rules for exponents work. For example, explain why $2^{-3} = \frac{1}{2^3}$, since $\frac{2^1}{2^4} = \frac{2}{2 \cdot 2 \cdot 2 \cdot 2} = \frac{1}{2^3}$ and $\frac{2^1}{2^4} = 2^{1-4} = 2^{-3}$.
Limitations	◦ Limit exponents to integers.
A1.4.2: Distinguish between situations that can be modeled with linear and exponential functions.	
Clarifications	a. Provide example data sets to explore possible linear or exponential relationships using technology. b. Data sets with no correlation are allowed.
Limitations	◦ Use simple, clean patterns where the data clearly exhibits a constant rate (linear) or constant ratio (exponential).
A1.4.3: Choose and fit an exponential function to given data using technology and justify the selected model based on the quantities in the given context; use that model to solve problems in the context of the data.	
Clarifications	a. Provide example data sets to explore possible linear or exponential relationships of the form $f(x) = a \cdot b^x$ using technology. b. Interpret the values of a and b in the equation $f(x) = a \cdot b^x$ in context.
Limitations	◦ Functions should be of the form $f(x) = a \cdot b^x$ with no transformations of this function (no reflections or translations).

A1.4.4: Given a context or graph of an exponential relationship between two variables, determine and interpret the key features and write an equation for the exponential function.	
Clarifications	a. Write equations in the form $y = a \cdot b^x$ where a = initial value (when $x = 0$) and b = base. b. Key features include the initial value (y-intercept), growth decay factor (base of the exponent) and end behavior.
Limitations	◦ Avoid exponents with coefficients (example: $y = a \cdot b^{2x}$). ◦ When given a graph, consecutive points should be identified. ◦ Restrict data to regular intervals (example: hours, months, years) and whole number exponents.
A1.4.5: Graph exponential functions expressed symbolically and show key features of the graph, by hand or using technology for more complicated cases. Interpret key features in terms of quantities in a context.	
Clarifications	a. The equation for the exponential function should be given, and should be in the form $f(x) = a \cdot b^x$. b. Key features include y-intercept, common factor (i.e., ratio or rate of growth/decay), domain, range, and end behavior.
Limitations	None
A1.4.6: Given two ordered pairs in an exponential relationship context, write an algebraic function to represent that relationship.	
Clarifications	a. Determine the growth/decay factor from two points obtained from a graph, table, or context. Equations for functions are explicit in the form $f(x) = a \cdot b^x$. b. The input values should be consecutive and all coordinates provided should be rational numbers.
Limitations	◦ Does not include the symbolic form for recursive form: e.g., $f(0) = 2$ and $f(x) = 3 \cdot f(x - 1)$.
A1.4.7: Solve exponential equations, by hand in simple cases, or with technology for more complicated cases.	
Clarifications	a. Equations that are to be solved by hand should be of a form that can be reduced to a form $b^{g(x)} = b^n$ where $g(x)$ is a linear function of x and n is an integer.
Limitations	◦ Does not require the symbolic form for recursive form (e.g., $f(0) = 2$ and $f(x) = 3 \cdot f(x - 1)$). ◦ This should not include using logarithms to solve an exponential equation, nor require converting radicals to rational exponents. ◦ Exponential equations should have the variable in the exponent on one side of the equation.

ʻŌpaʻa 5 - Exploring Quadratic Functions & Equations

Students deepen their understanding of functions by studying quadratic functions. They analyze key features of graphs to describe the behavior of these functions and explore operations with polynomials—adding, subtracting, multiplying, and factoring polynomials of degree no more than two. Students solve quadratic equations using graphs, factoring, and the quadratic formula (limited to real solutions), and graph quadratic functions to strengthen their understanding of quadratic structure and behavior. Given data, students use technology to create quadratic functions and use those functions to estimate solutions to problems.	
A1.5.1: Choose and fit a quadratic function to given data using technology and justify the selected model based on the quantities in the given context; use that model to solve problems in the context of the data.	
Clarifications	a. Provide example data sets to explore possible quadratic relationships using technology. Domain and range should be considered in terms of context.
Limitations	None
A1.5.2: Given a graph that models a quadratic relationship between two quantities, use key features to interpret the graphs in terms of the quantities.	
Clarifications	a. Key features include x-intercept(s), y-intercept, domain, range, vertex, maximum/minimum value, axis of symmetry.
Limitations	None
A1.5.3: Graph quadratic functions and show key features of the graph. Interpret key features in terms of quantities in a context.	
Clarifications	a. Graph quadratic functions given in various forms (standard, vertex, factored). Utilize a table to locate the vertex, then symmetry to complete the graph. Observe the following: ◦ In standard form $f(x) = ax^2 + bx + c$, the value of c gives the y-intercept of the graph of $y = f(x)$. ◦ In vertex form $f(x) = a(x - h)^2 + k$, (h, k) gives the coordinates of the vertex of the graph of $y = f(x)$. ◦ In factored form $f(x) = a(x - r_1)(x - r_2)$, r_1 and r_2 give the x-intercepts of the graph of $y = f(x)$. ◦ In all three forms, a determines the direction the parabola opens. b. Graph similar quadratic functions in each of the different forms without graphing technology (i.e., creating tables of values and plotting points). c. Technology should be used to graph more complicated quadratic functions and to help with making generalizations about the various forms.
Limitations	◦ Does not need to use transformations to explore quadratic functions. ◦ Quadratic functions should have vertices with x-values in the interval $[-5, 5]$.

A1.5.4: Select an equivalent form of a quadratic function to reveal properties of the quantities represented by the equation.	
Clarifications	a. From a given set of two or three forms (standard, factored, vertex), select the form(s) that immediately reveal certain key features of the quadratic function without any additional work or computations. ◦ Standard form reveals the y-intercept. ◦ Factored form reveals the x-intercepts. ◦ Vertex form reveals the axis and vertex. ◦ All three forms reveal the direction of the graph, and whether or not the function has a maximum or minimum.
Limitations	◦ Does not need to rewrite the equation of a quadratic function in the different forms themselves.
A1.5.5: Add, subtract, and multiply polynomials that result in polynomials of degree less than or equal to two.	
Clarifications	a. Include difference of squares and perfect square trinomials. b. Write answers in simplest form (i.e., like terms are combined, and terms are written in descending order). c. As enrichment, sums/differences/products that result in polynomials of degree greater than two may be explored.
Limitations	None
A1.5.6: Factor simple polynomial expressions.	
Clarifications	a. Factoring limited to polynomials of degree 4 or less: ◦ Greatest common factor of any polynomial. ◦ Differences of squares with one or two variables, leading coefficient can be any perfect square. ◦ Trinomials with a quadratic leading coefficient of one. ◦ Polynomial expressions that, after factoring out a greatest common factor, results in a trinomial with a quadratic leading coefficient of one.
Limitations	◦ Not required to factor the following: ◦ Sum/difference of cubes. ◦ Quadratic trinomials with quadratic leading coefficients other than one.

A1.5.7: Solve quadratic equations with real solutions.	
Clarifications	a. Methods for solving quadratics should include all of the following: ◦ By algebraic manipulation if they equation can be rewritten as $x^2 = k$. ◦ By algebraic manipulation using factoring and the zero-product property when the leading coefficient is one. ◦ Using the quadratic formula (the formula is provided).
Limitations	◦ When using the quadratic formula, with real solutions (the formula is provided): ◦ Rewrite expressions so there is a single value in the radicand and in the denominator: E.g., For the equation $x^2 + 4x - 6 = 0$, the solution $x = \frac{-4 \pm \sqrt{4^2 - 4(1)(-6)}}{2(1)}$ should be rewritten to $x = \frac{-4 \pm \sqrt{40}}{2}$. Any further equivalent forms (e.g., $x = \frac{-4 \pm 2\sqrt{10}}{2} = -2 \pm \sqrt{10}$) following from this would also be acceptable, but not required. ◦ Find approximate values for irrational solutions: E.g., $x = \frac{-4 \pm \sqrt{40}}{2}$ is $x \approx 1.16$ or $x \approx -5.16$
A1.5.8: Create contextual equations in one variable and use them to solve problems.	
Clarifications	None
Limitations	◦ Problems should be clear and focused, ensuring the relevant quantities and relationships needed to form a single-variable quadratic equation are readily identifiable. ◦ Equations should yield real number solutions.

ʻŌpaʻa 6 - Statistical Modeling with Regression Analysis

Students are able to synthesize concepts found in all of the other big ideas to make sense of real-world data. Students will work with different types of data—linear, exponential, and quadratic—and develop mathematical models to represent them. They demonstrate a deep understanding of how functions help us describe, analyze, and predict real-world phenomena, and explain and justify their reasoning behind their models.	
A1.6.1: Choose and fit a linear, exponential, or quadratic function to given data using technology and justify the selected model based on the quantities in the given context.	
Clarifications	a. Use technology to compute the correlation coefficients of linear, exponential, or quadratic fits. Compare the correlation coefficients to choose a model.
Limitations	None

A1.6.2: Use a model for a given data set to solve problems in the context of the data.	
Clarifications	a. This standard builds from the data or scenario used in A1.6.1 to dive deeper into the analysis of the model.
Limitations	None
A1.6.3: For a function that models a linear, exponential, or quadratic relationship between two quantities, use key features to interpret the graphs and tables in terms of the quantities.	
Clarifications	a. This standard may build upon the data or scenario used in A1.6.1 and A1.6.2. b. Key features include: intercepts, intervals where the function is increasing, decreasing, positive, or negative, relative maximums and minimums; symmetries; and end behavior.
Limitations	None
A1.6.4: Interpret expressions that represent a quantity in terms of its context such as terms, constants, coefficients, exponents.	
Clarifications	a. This standard may build upon the data or scenario used in A1.6.1 and A1.6.2.
Limitations	None
A1.6.5: Compare properties of two functions each represented in a different way (algebraically, graphically, numerically in tables, or by verbal descriptions).	
Clarifications	None
Limitations	None
A1.6.6: Represent constraints by equations or inequalities and interpret solutions as viable or non-viable options in a modeling context.	
Clarifications	a. Communicate restrictions on the domain of a fitted model based on context.
Limitations	None
A1.6.7: Calculate, interpret, and apply the average rate of change of a function (over a specified interval) to estimate other values.	
Clarifications	a. Use linear, exponential, and quadratic models.
Limitations	None

Algebra 2

ʻŌpaʻa 1 - Modeling Single Relationships

Students extend their earlier work with functions from Algebra 1 + Data Analytics through the use of technology to model situations with linear, quadratic, and exponential functions and examine the appropriateness of these models using correlation coefficients (r) and coefficients of determination (r^2). They justify the appropriateness of the model based on the meanings of the quantities and how they change, focus on the importance of domain and range, and use technology to examine the fit of the model, including articulating cautions about extending the model beyond its initial bounds. Students translate contextual scenarios into piecewise functions, including the absolute value function, and analyze, evaluate, and graph these functions from their algebraic representations.

A2.1.1: Use technology to fit linear, quadratic, or exponential functions to data and apply these models to solve problems in context.	
Clarifications	None
Limitations	◦ Evaluation and analysis is based on graphical, numerical, and verbal representations, not algebraic (which will be addressed later in the course).

A2.1.2: Find the correlation coefficient, r, and the coefficient of determination, r^2, using technology, and use them to interpret the appropriateness of a given model.	
Clarifications	a. Use of larger data sets is encouraged, but not required.
Limitations	◦ Do not calculate r and r^2 algebraically.

A2.1.3: Explain cautions and qualifiers when extrapolation is necessary or discussed.	
Clarifications	None
Limitations	None

A2.1.4: Identify the domain and range of a linear, quadratic, or exponential function, with and without contextual limits.	
Clarifications	a. For a given context, domain and/or range can differ between the data representation and the model determined by the data. For example, consider discrete versus continuous sets of values.
Limitations	◦ Domain and range should be described in words or inequalities.

A2.1.5: Model graphically a contextual scenario using piecewise linear functions.	
Clarifications	None
Limitations	◦ Piecewise linear functions should have no more than 3 different parts.
A2.1.6: Evaluate and graph piecewise functions, including absolute value, from algebraic representations.	
Clarifications	a. When evaluating piecewise functions, include function types other than linear pieces, such as polynomial, rational, or exponential pieces. b. Absolute value functions should be rewritten as a piecewise linear function with two parts (transformations standards appear later).
Limitations	◦ When evaluating, piecewise functions should have no more than 3 different parts. ◦ Absolute value functions should be of the form $f(x) = \|mx + b\|$ or $f(x) = \|mx\| + b$.

ʻŌpaʻa 2 - Modeling Multiple Relationships

A natural extension from working with one relationship between two varying quantities is to examine two relationships simultaneously. Students make sense of real-world situations and translate those into pairs of equations with two variables and show flexibility in finding solution pairs that solve both equations simultaneously. They solve systems using substitution, elimination, and technology, and make sense of those solutions in terms of the situation.	
A2.2.1: Model a contextual scenario using two linear equations in two variables.	
Clarifications	a. Modeling should be in context, creating two equations that occur simultaneously.
Limitations	None
A2.2.2: Given a system of two linear equations in two variables, use the method of substitution, elimination, or graphing technology to solve the system.	
Clarifications	a. Equations may be given in any form (standard, slope-intercept, etc.).
Limitations	None

A2.2.3: Solve a simple system consisting of a linear equation and a quadratic equation in two variables algebraically and using graphing technology.	
Clarifications	a. When finding a solution, prior understanding of either form of the quadratic formula is applied: $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$ or $x = \frac{-b}{2a} \pm \frac{\sqrt{b^2-4ac}}{2a}$.
Limitations	◦ Both equations should be given explicitly (i.e., $y =$ an expression in terms of x). ◦ When solving algebraically, the resulting quadratic equation should be factorable with leading coefficient 1. ◦ Further exploration of quadratic equations is addressed in related standards.

ʻŌpaʻa 3 - Manipulating Exponential Expressions & Equations

Students rewrite expressions with rational exponents, reasoning with and applying exponent properties to simplify and transform them. They exhibit representational flexibility as they convert between radical and fractional exponent forms, and use algebraic skills and technology to solve equations of the form $a \cdot x^b = k$, $a \cdot b^x = k$, and $b^x = c^{dx}$. Students solve exponential equations in different ways, including by recognizing or creating a common base. Then more complex equations are provided that warrant the use of technology, justifying the need for other mathematical techniques in subsequent standards.

A2.3.1: Rewrite radical expressions using rational exponents and vice versa.	
Clarifications	None
Limitations	◦ Radical and rational expressions must have no more than 2 different variables.

A2.3.2: Rewrite expressions involving radicals and rational exponents using the properties of exponents.	
Clarifications	None
Limitations	◦ Expressions must have no more than 2 different variables. ◦ Sums and differences of numerical radical expressions should be no more than 2 terms.

A2.3.3: Solve power equations of the form $a \cdot x^b = k$ algebraically.	
Clarifications	None
Limitations	◦ Equations involving power functions should be of the form $a \cdot x^b = k$, where: ◦ $b = \frac{1}{2}, \frac{1}{3}, 2,$ or 3 ◦ a and k are integers ◦ $\frac{k}{a}$ is an integer ◦ Solutions are real ◦ Solving using technology will be covered in subsequent standards.

A2.3.4: Solve exponential equations of the form $a \cdot b^x = k$ and $b^x = c^{dx}$ algebraically.	
Clarifications	None
Limitations	◦ For equations of the form $a \cdot b^x = k$, $\frac{k}{a}$ should be an integer power of b. ◦ For equations of the form $b^x = c^{dx}$, the values of b and c^d should be integer powers of a common base. ◦ Exploration of logarithms is addressed in related standards.
A2.3.5: Solve radical, power, and exponential equations using graphing technology.	
Clarifications	None
Limitations	None

ʻŌpaʻa 4 - Understanding Inverse Functions & Equations

Students understand an inverse function as another way of representing a given function situation, but one that reverses the input-output process. They identify when a relationship is an inverse of another function from graphs, equations, or contexts, and find and interpret inverse function values. They can explain and use the inverse relationship between exponential and logarithmic functions, and use this connection to solve problems involving both. Students analyze and graph logarithmic functions, paying close attention to domain, intercepts, and end behavior. Students use function composition in contextual and mathematical situations and use composition to verify inverse relationships algebraically.	
A2.4.1: Given a graph or equation of a contextual linear function, identify the input value corresponding to a particular output value.	
Clarifications	None
Limitations	None
A2.4.2: Read and interpret values of an inverse function from a graph or a table, given that the function has an inverse.	
Clarifications	a. Inverse function notation is introduced here. Initially, function and inverse function notation should be analyzed using contextual situations. This sets the foundation for later analyzing these notations in non-contextual situations.
Limitations	None

A2.4.3: Rewrite exponential equations of the form $b^x = k$ and $y = b^x$ as logarithmic equations.	
Clarifications	a. Promote the idea that logarithms are exponents. b. Introduce common logarithms (base 10).
Limitations	◦ b should be a positive integer greater than 1. ◦ For equations of the form $b^x = k$, k should not be a rational power of b. ◦ Further exploration with logarithms is addressed in Precalculus with Trigonometry and AP Precalculus.
A2.4.4: Graph logarithmic functions with technology, showing intercepts and end behavior.	
Clarifications	a. Interpreting intercepts and end behavior in contextual situations is encouraged, but not required.
Limitations	◦ Further exploration with logarithms is addressed in Precalculus with Trigonometry and AP Precalculus.
A2.4.5: Express the solution to exponential equations of the form $a \cdot b^{cx} = d$ as a logarithm, evaluating the logarithm using technology.	
Clarifications	a. Exponential equations should be explored in both purely mathematical and contextual situations. b. Introduce natural logarithms (base e) in context.
Limitations	None

ʻŌpaʻa 5 - Exploring Quadratic Functions & Equations

Students extend earlier understandings of the applications, behavior and manipulation of quadratic functions. They examine quadratic relationships in terms of inputs and outputs and think about how changes in one variable result in changes to the other. They graph quadratic functions by analyzing transformations of the basic function $f(x) = x^2$ and interpret meanings of those transformations in terms of the quantities in the situation, and conversely, derive quadratic function equations from given graphs. Students apply quadratic functions to model real-world scenarios, translating contextual information into mathematical representations. Students flexibly solve quadratic equations algebraically using the quadratic formula and factoring, and make sense of solutions in terms of contexts. They use the discriminant to determine the number of real or complex solutions.	
A2.5.1: Graph quadratic functions resulting from at most two transformations of $f(x) = x^2$, given a verbal description of the transformations.	
Clarifications	a. Transformations include vertical and horizontal shifts, vertical dilations (widening and narrowing), and reflection across the x-axis. b. Quadratic functions resulting from shifts and/or reflections should be graphed without technology. c. Quadratic functions involving vertical dilations should be graphed with technology.
Limitations	◦ Transformation of functions should be limited to one or two transformations at a time.
A2.5.2: Write a quadratic function in standard form, given its graph.	
Clarifications	None
Limitations	◦ The quadratic function should be based on at most two transformations of $f(x) = x^2$.
A2.5.3: Create a quadratic function in vertex form to model an appropriate contextual scenario, given a verbal description. Interpret the parameters of the function in terms of the context.	
Clarifications	a. Verbal descriptions might include scenarios involving projectiles. b. When interpreting parameters, address whether there is a relative maximum or minimum (sign of a), and the meaning of the vertex in context (values of h and k).
Limitations	◦ Apply only to $f(x) = x^2$, which yields $f(x) = a(x - h)^2 + k$.
A2.5.4: Determine the discriminant of a quadratic expression in standard form to reveal the number of real zeros of the function it defines.	
Clarifications	None
Limitations	None

A2.5.5: Use the quadratic formula to solve quadratic equations with integer coefficients that have complex solutions. Write the complex solutions in the form $a + bi$ for real numbers a and b.	
Clarifications	a. Use the definition of the imaginary unit i (i.e., $i^2 = -1$) to rewrite square roots with negative integer radicands. b. When rewriting square roots, the radical expression should be rewritten so that the radicand does not contain perfect square factors.
Limitations	◦ When using the quadratic formula, the resulting radicand should be a negative integer.
A2.5.6: Factor quadratic expressions with leading coefficient other than 1.	
Clarifications	None
Limitations	◦ Quadratic expressions should be factorable over the integers with integer coefficients. ◦ Quadratic expressions should be created to allow for factoring within a reasonable number of attempts, preventing excessive time spent on random guessing. ◦ Factors should be linear terms with coefficients of 7 or less (i.e., $ax \pm b$ with $1 \leq a, b \leq 7$).

ʻŌpaʻa 6 - Investigating Polynomial Functions & Equations

Students investigate and make sense of the applications, behavior and manipulation of polynomial functions. They examine polynomial relationships in terms of inputs and outputs and think about how changes in one variable result in changes to the other. They can explain the connection between factors, zeros, and graphical characteristics and identify real and complex roots from factored forms. They analyze end behavior from algebraic forms, and explain how polynomials of different degrees behave at extreme values. Utilizing technology, students graph polynomial functions, identifying real zeros, intervals of increase and decrease, and relative extrema. Students exhibit representational flexibility by using algebraic manipulations to rewrite polynomials in equivalent forms, including applying operations involving polynomial arithmetic, focusing on addition, subtraction, and multiplication, with results limited to degrees 3-5. A deeper exploration of mathematical models developed earlier in the course provides context for these mathematical concepts.

A2.6.1: Given a polynomial function in factored form, determine the real and complex zeros of the function.	
Clarifications	None
Limitations	◦ Polynomial functions given in factored form should be products of linear and irreducible quadratic factors. ◦ The zeros of the polynomial function should have multiplicity at most 1.

A2.6.2: Determine the degree and sign of the leading coefficient for a polynomial expression; use covariational language to informally describe the end behavior of the function it defines.	
Clarifications	a. Covariational language refers to the way we describe and reason about how two or more quantities change together. For example: "As x becomes very large and positive, $f(x)$ becomes very large and negative."
Limitations	◦ Technology should be used when graphing.
A2.6.3: Create a polynomial function to model an appropriate contextual scenario, given a verbal description. Use technology to estimate: (a) the function's real zeros; (b) intervals where the function is increasing or decreasing; and (c) relative maximums and minimums.	
Clarifications	a. Verbal descriptions can include applications (e.g., volume of boxes with side lengths). b. Since polynomial functions are continuous, when stating intervals where a function is increasing or decreasing, the response should include any endpoints (e.g., the function $f(x) = x^3 - 3x$ is decreasing on the interval $-1 \leq x \leq 1$). c. When stating relative maximums and minimums, the response should involve the output value, not the input value or coordinate pair on a graph (e.g., the function $f(x) = x^3 - 3x$ has relative maximum 2).
Limitations	◦ The zeros of the polynomial function should have multiplicity at most 1.
A2.6.4: Add, subtract, and multiply two polynomials, with results limited to degree 3-5.	
Clarifications	None
Limitations	◦ In Algebra 1 + Data Analytics, polynomial addition, subtraction, and multiplication are limited to results of degree 0-2.

ʻŌpaʻa 7 - Introducing Rational Functions

Students investigate and make sense of the applications, behavior and manipulation of simple rational functions. They examine those relationships in terms of inputs and outputs and think about how changes in one variable result in changes to the other. Students use both context and algebra to establish the function's domain and identify vertical asymptotes, and demonstrate insight into the restrictions and behaviors of these functions. Students apply rational functions as models of real-world scenarios, strengthening their understanding of these functions, including inversely proportional relationships.	
A2.7.1: Determine the domain of a simple rational function and the vertical asymptote of its corresponding graph.	
Clarifications	a. When stating the domain, responses should be presented in both exclusionary (e.g., $x \neq 2$) and inequality (e.g., $x < 2$ or $x > 2$) forms. b. When stating the vertical asymptote, an equation of the form $x = h$ should be used, rather than a single number corresponding to the x-value of the vertical asymptote.
Limitations	◦ A simple rational function is a non-zero function where the numerator is constant or linear, the denominator is linear, and the numerator and denominator have no common factors.
A2.7.2: Create a simple rational function to model an appropriate contextual scenario, given a verbal description.	
Clarifications	a. Contexts can include applications such as rate problems, average profit, elasticity of demand, etc.
Limitations	◦ A simple rational function is a non-zero function where the numerator is constant or linear, the denominator is linear, and the numerator and denominator have no common factors.

Quantitative Reasoning

ʻŌpaʻa 1 - Developing Number Sense and Problem-Solving Skills

Students develop strong number sense by learning to evaluate the reasonableness of solutions in both real-world and mathematical problems using context clues and estimation strategies. Through exploring ratios, proportions, percentages, and unit rates, they learn to make informed, practical decisions in complex situations. Converting units within and between measurement systems strengthens their ability to navigate diverse problem types with precision. By applying the properties and operations of various number systems—including understanding place value—they deepen their conceptual understanding of how numbers function across different contexts. This foundational number sense empowers students to think flexibly, reason logically, and apply mathematics meaningfully in their daily lives. Students solve real world problems catered to the career interests of students (nurses: dosage calculations, firefighters: fire burn rates, construction: scale drawings/conversions, etc.)	
QR.1.1: Assess the validity and reasonableness of solutions by using estimation strategies and comparing magnitudes of quantities within real-world and mathematical contexts.	
Clarifications	a. Engage with problems that involve different constraints, scenarios, and variables to introduce multiple equations and bounds on solutions. b. Develop mathematical literacy by reading and interpreting word problems and solutions with understanding and accuracy. c. Use estimation as a strategic tool to verify reasonableness and to support mental math and decision-making. d. Evaluate the reasonableness, limitations, and assumptions of AI driven analyses and solutions and justify conclusions using mathematical evidence.
Limitations	None
QR.1.2: Use ratios, proportions, percentages, and unit rates to solve problems and make informed decisions in complex, real-world, mathematical situations.	
Clarifications	a. Move beyond basic computation into multi-step reasoning, abstract representation, and problem modeling. b. Justify decisions with quantitative reasoning, using solutions as evidence.
Limitations	None

QR.1.3: Use dimensional analysis to reason about and perform unit conversions within and between measurement systems, ensuring accuracy and coherence in problem-solving.	
Clarifications	a. Conversions are embedded in multi-step, multi-unit problems, and dimensional analysis in both the U.S. customary system and the metric system, as well as conversions within each system. b. Use dimensional analysis as a tool for reasoning, not just for converting. c. Elevate typical length/volume/mass units to include rates, densities, currency, and composite units.
Limitations	None

ʻŌpaʻa 2 - Analyzing Functions

Students will understand how to model, interpret, and compare linear and exponential functions using multiple representations—tables, graphs, equations, and contextual scenarios. They will recognize patterns of additive and multiplicative change, identify key features such as slope, intercepts, growth/decay rates, and asymptotes, and use these to analyze real-world phenomena. Students will also develop the ability to distinguish between linear and exponential models and evaluate which is appropriate for a given situation.	
QR.2.1: Use technology to develop linear models from data, interpret the meaning of model parameters in context, and use the model to make predictions and solve real-world problems.	
Clarifications	a. Recognize and construct linear functions from tables, graphs, equations, and real-world contexts. b. Determine the rate of change (slope) of a linear relationship and key values (intercepts) of the function both in and out of context. c. Translate between representations to reveal various properties of linear functions.
Limitations	None
QR.2.2: Use technology to construct exponential models from data, interpret model outputs and limitations in context, and use the model to make informed predictions and decisions.	
Clarifications	a. Recognize and construct exponential functions from tables, graphs, equations, and real-world contexts. b. Determine key parameters of an exponential function (growth/decay rate and y-intercept) both in and out of context. c. Identify and describe the asymptotic behavior of an exponential function.
Limitations	◦ Exponential functions should be given as $y = a \cdot b^x$ and should not include vertical transformations of these functions. ◦ Determining the closed form of an exponential function given two points (x_1, y_1) and (x_2, y_2) where $\lvert x_1 - x_2 \rvert = 1$ should be encouraged. Determining the closed form of a function given points where $\lvert x_1 - x_2 \rvert > 1$ is not required.

QR.2.3: Analyze and compare linear and exponential growth and decay in real-world contexts by calculating and interpreting absolute and relative change, and use these comparisons to justify conclusions.	
Clarifications	a. Analyze and compare how various linear and exponential functions grow or decline over time. b. Determine constant growth/decay rates and proportional growth and decay factors. c. Interpret the similarities and differences graphically, numerically, and contextually allowing for various representations to arise based on the context. d. Model situations that involve additive and/or multiplicative growth or decay.
Limitations	None
QR.2.4: Identify an appropriate model for a bivariate data set (e.g., linear, exponential) to approximate solutions for real world problems, identify patterns and identify how changing parameters affect the models.	
Clarifications	a. Use of larger data sets is encouraged, but not required. b. Appropriate models may be identified by calculating the correlation coefficient and coefficient of determination.
Limitations	◦ Not expected to calculate r and r^2 without the use of technology.

ʻŌpaʻa 3 - Utilizing Financial Math for Decision Making

Students understand how interest, investment, and debt function within personal finance contexts. They apply mathematical reasoning to make informed decisions about saving, borrowing, and investing. Students recognize the long-term financial impact of interest rates, payment strategies, and investment options, and they evaluate and compare various financial scenarios to make sound financial choices.	
QR.3.1: Apply appropriate mathematical models to analyze various types of loans. Determine the impact of changes to loan rates, terms, and payments, including making additional payments toward the principal.	
Clarifications	a. Apply the simple interest formula $I = Prt$ and compound interest formula $A = P(1 + \frac{r}{n})^{nt}$. b. Include comparisons of short-term and long-term loans to evaluate how time affects total interest paid. c. Use realistic contexts such as student loans, car loans, and mortgages. d. Analyze how early or additional payments reduce total interest paid and shorten the loan duration. e. Include exploration of amortization and how payments are split between principal and interest over time. f. Interpret the benefit of paying off high-interest debt first.

Limitations	◦ Do not introduce logarithmic functions or rules of logs. ◦ Interest calculations should be limited to annual, quarterly, or monthly compounding. ◦ Full amortization schedules are not required, though simple tables and comparisons of a few sample months may be used. ◦ Calculators may be used for arithmetic, but not for generating entire amortization charts.
QR.3.2: Identify and distinguish between types of credit and debt, and analyze their long-term financial implications.	
Clarifications	a. Include types such as credit cards, personal loans, student loans, car loans, and mortgages. b. Explore concepts like interest rates, minimum payments, fees, and how credit affects financial decisions. c. Emphasis should be placed on understanding credit reports and how debt impacts financial health over time.
Limitations	◦ Interpreting actual credit reports or calculating credit scores is not required. ◦ Focus on understanding over precise technical details of financial regulations.
QR.3.3: Calculate and interpret the return on investment (ROI) for various financial products and investment strategies.	
Clarifications	a. Use the ROI formula: $\mathrm{ROI} = \frac{\text{gain-cost}}{\text{cost}}$. b. Contextual scenarios might include evaluating profit from stock investments, savings accounts, or real estate. c. Include both percentage and dollar interpretations of ROI.
Limitations	◦ Exclude considerations like inflation, tax implications, or fees unless clearly given in the context. ◦ Modeling fluctuating market returns or long-term investment portfolios is not required.
QR.3.4: Analyze and compare the characteristics of various investment options, including return rates, risks, and time horizons.	
Clarifications	a. Compare investment products such as savings accounts, CDs, and bonds using the concepts of risk vs. reward, compounding frequency, and time horizon. b. Emphasis should be placed on comparing annual percentage yield (APY) versus annual percentage rate (APR), and how compounding affects returns. c. Technology should be used for calculations. d. Analyze and compare the effects of different compounding scenarios on an investment of equal initial value, using mental calculation, estimation, or reasoning rather than technology.
Limitations	◦ Calculations involving continuous compounding or advanced financial instruments (e.g., derivatives) are not required.

ʻŌpaʻa 4 - Exploring and Analyzing Descriptive Statistics

Through analyzing data, modeling patterns, and exploring distributions and relationships, students will be able to gain essential knowledge that drives informed decision-making. Students will explore graphs and data in the real world and be asked questions like "what do you wonder?" or "what's going on in this graph?". As students wonder about data, they will develop a deeper understanding of information that is conveyed. Students will gather reliable, unbiased data and make valid conclusions about a larger population. Students will use technology to construct graphical representations of numerical data.

QR.4.1: Analyze and interpret data by selecting and justifying the appropriate measure(s) of central tendency.

Clarifications	a. For quantitative data: ◦ Calculate and interpret mean, median, and mode, and determine which measure is most appropriate based on the shape of the data distribution. ◦ Understand how outliers and skewness affect each measure. ◦ Interpretations should include contextual reasoning (e.g., explaining why the median is a better measure of center in a skewed distribution). ◦ Calculate weighted and trimmed means and explain what these measures tell us about the data, and decide when each is the most meaningful way to represent a data set. b. For qualitative data: ◦ Understand why some measures of central tendency are not appropriate for a qualitative data set. c. For both: ◦ Data should be presented in list or graphical form. Analysis should be performed from either starting point.
Limitations	◦ Calculus-based derivation or inference is not required.

QR.4.2: Analyze and interpret the variability of a data set by selecting and justifying the appropriate measure(s) of spread.

Clarifications	a. Calculate and interpret range and standard deviation and understand what these measures say about consistency and variability in data. b. Standard deviation should be conceptually understood as the average distance from the mean. c. If calculating standard deviation by hand, do so only for data sets < 5 for the purpose of understanding where the value comes from. d. Visual displays (especially box plots) should be used to support interpretation of spread. e. Allow for multiple methods for defining outliers, allowing an analysis of how outliers impact variability and select appropriate measures of spread.
Limitations	◦ The derivation of standard deviation formulas is not required. ◦ Variance may be introduced for context but not emphasized in calculation.

QR.4.3: Distinguish between quantitative and qualitative data and determine appropriate methods of analysis and visual representation based on the nature of the data.	
Clarifications	a. Identify whether a variable is quantitative (numerical) or qualitative (categorical). b. Describe how to summarize and visualize each type appropriately (e.g., bar charts for categorical data and dot plots, histograms, box plots for numerical). c. Interpret features such as center, spread, shape (symmetry, skewness), and the presence of outliers in visual representations (e.g., dot plots, histograms, box plots). d. Discussions should include which statistical measures (e.g., mean, median, mode) are applicable to each data type. e. Data sets should range in size (<10 to >100) so the best type of visualization can be determined for different scenarios.
Limitations	◦ The use of statistical computing software (e.g., R or Python) is not expected, though calculators and/or spreadsheets may be used. ◦ Focus remains on univariate data; analysis of multivariate data is not required. ◦ Choosing appropriate graph types should not be dependent on shape or measures of spread. ◦ Inferential methods (e.g., hypothesis testing for proportions or means) are not required.

ʻŌpaʻa 5 - Calculating and Interpreting Probability

Students will understand the likelihood of outcomes and the foundation for making inferences from data. Students will develop an idea of randomness and uncertainty in data, quantify the likelihood of events, and make connections between independence, mutual exclusivity, and conditional probability. Students will utilize visual and numerical representations to understand relationships among data. They will calculate expected value and evaluate its meaning in real life and its role in decision making.	
QR.5.1: Apply fundamental set-theoretic concepts to represent, analyze, and interpret relationships among sets and events using symbolic notation, tables, and quantitative representations, and use these relationships to draw justified conclusions.	
Clarifications	a. Analyze and apply set relationships (unions, intersections, complements, and subsets) using set notation, Venn diagrams, and contextual examples. b. Use two-way or frequency tables to organize categorical data and analyze relationships between two variables.
Limitations	None

QR.5.2: Calculate probabilities, including conditional probabilities, and use probabilities to inform decision making.	
Clarifications	a. Understand that probability measures the likelihood of events as the ratio of favorable outcomes to total equally likely outcomes, and be able to calculate probabilities using either theoretical sample spaces or observed data. b. Understand that independent events do not affect each other's outcomes, and mutually exclusive events cannot occur at the same time. c. A compound event occurs when two or more events are considered together, with "or" representing the union of events and "and" representing the intersection of events. d. Interpret and communicate the meaning of a probability in context, including its long-run or relative-frequency interpretation (e.g., a probability of 0.2 corresponds to the event occurring approximately 20% of the time over many trials). e. Use tools (i.e. Venn diagrams, two-way tables, tree diagrams) to support analysis. f. Design and conduct simulations to approximate probabilities, interpret the outcomes, and use them to guide decisions in uncertain situations.
Limitations	◦ Events should involve finite, countable outcomes (e.g., dice, spinners, card draws) rather than continuous probability distributions. ◦ Formal proofs or advanced symbolic probability notation are not required for probability. ◦ Bayes' Theorem is not required.
QR.5.3: Compute expected value and analyze variability of a random variable to draw conclusions about mathematical fairness, payoff, and risk.	
Clarifications	a. Expected value should be introduced as a weighted average of outcomes, calculated as the sum of $(\text{value} \times \text{probability})$ in the context of repeated trials or decision-making. b. Recognize that expected value represents a long-run average, not a guaranteed result. c. Use expected value to evaluate whether a game is "fair" (i.e., expected value is zero) or to compare different options in a decision. d. Discussions should connect mathematical results to real-life implications (e.g., risk, utility, long-term gain or loss) using real-world scenarios like games of chance, insurance, investments, contests, or GPA.
Limitations	◦ Focus should remain on discrete outcomes and probabilities (e.g., games, payouts, lottery scenarios). ◦ No calculus or integration required for expected value of continuous distributions.

ʻŌpaʻa 6 - Modeling with Trigonometric Relationships

Students will understand how to apply trigonometric relationships in triangles to solve real-world and mathematical problems involving distances and angle measures. They will use sine, cosine, and tangent ratios, as well as the Laws of Sines and Cosines, to model scenarios, interpret results, and evaluate the reasonableness of their solutions using both numeric and algebraic methods with and without technology.	
QR.6.1: Apply right-triangle trigonometric relationships to construct models and solve real-world and mathematical problems involving distances and angle measures.	
Clarifications	a. Use sine, cosine, and tangent ratios to solve for missing side lengths or angles in right triangles. b. Problems should include real-world applications, such as: ◦ Measuring height using shadows or angles of elevation/depression ◦ Finding distances across rivers or inaccessible areas ◦ Modeling ramps, ladders, and slopes ◦ Construction related problems c. Right triangle scenarios should be visually represented through diagrams or created from word problems. d. Both numeric (e.g., sides or angles given as numbers) and algebraic (e.g., solve for a variable) problem types should be included. e. Make use of technology (e.g., calculators or digital tools) when appropriate, especially for inverse trigonometric functions.
Limitations	◦ All angles should be given and measured in degrees. The use of radian measures and conversions between degrees and radians are not required.
QR.6.2: Understand when and how to apply the Law of Sines and the Law of Cosines to solve real-world and mathematical problems including finding unknown measurements in right and non-right triangles.	
Clarifications	a. Determine what situations will require the law of sines, law of cosines, or trigonometric ratios.
Limitations	◦ A proof or derivation of the law of sines and cosines is not required. ◦ Only direct applications of a single trigonometric law (sines or cosines) are appropriate. Tasks involving the integration of multiple laws are not required.